Algorithm Handbook

Continue

2026 Edition

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1 代数结构

1.1 幺半群

```cpp
template <typename Mono>
struct RevMono {
    using Type = typename Mono::Type;
    static Type e() {
        return Mono::e();
    }
    static Type op(const Type& a, const Type& b) {
        return Mono::op(b, a);
    }
};

template <typename T>
struct AddMono {
    using Type = T;
```

```
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {a + b};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type inv(const Type& a) {
        return {-a};
    }
    static Type pow(const Type& a, i64 k) {
        return a * k;
    }
};

template <typename T>
struct MulMono {
    using Type = T;
    static Type e() {
        return {1};
    }
    static Type op(const Type& a, const Type& b) {
        return {a * b};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type inv(const Type& a) {
        return T{1} / a;
    }
};

template <typename T>
struct MinMono {
    using Type = T;
    static Type e() {
        return {inf<T>};
    }
    static Type op(const Type& a, const Type& b) {
        return {min(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct MaxMono {
    using Type = T;
    static Type e() {
        return {-inf<T>};
    }
    static Type op(const Type& a, const Type& b) {
        return {max(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct GcdMono {
    using Type = T;
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {gcd(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct LcmMono {
    using Type = T;
```

```
    static Type e() {
        return {1};
    }
    static Type op(const Type& a, const Type& b) {
        return {lcm(a, b)};
    }
    static Type from(const i64& x) {
        return {x};
    }
    static Type pow(const Type& a, i64 k) {
        return (k == 0 ? e() : a);
    }
};

template <typename T>
struct XorMono {
    using Type = T;
    static Type e() {
        return {0};
    }
    static Type op(const Type& a, const Type& b) {
        return {a ^ b};
    }
    static Type inv(const Type& a) {
        return {a};
    }
    static Type power(const Type& a, i64 k) {
        return (k & 1) ? a : e();
    }
};
```

1.2 幺半群作用

```
// Template
// struct ActMono {
//     using Mono = Mono;
//     using Tag = Tag;
//     using X = Mono::Type;
//     using Y = Tag::Type;
//     static X act(const X& x, const Y& y, int len) {
//         return {};
//     }
// };

template <typename T>
struct SumAdd {
    using Mono = AddMono<T>;
    using Tag = AddMono<T>;
    using X =  Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y * len};
    }
};

template <typename T>
struct SumMul {
    using Mono = AddMono<T>;
    using Tag = MulMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x * y};
    }
};

template <typename T>
struct MinAdd {
    using Mono = MinMono<T>;
    using Tag = AddMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y};
    }
};

template <typename T>
struct MaxAdd {
    using Mono = MaxMono<T>;
    using Tag = AddMono<T>;
    using X = Mono::Type;
    using Y = Tag::Type;
    static X act(const X& x, const Y& y, int len) {
        return {x + y};
    }
};
```

1.3 半群快速幂

```
template <typename Mono>
```

```
typename Mono::Type power(typename Mono::Type a, i64 b) {
    if constexpr (requires { Mono::pow(a, b); }) {
        return Mono::pow(a, b);
    } else {
        auto res = Mono::e();
        while (b > 0) {
            if (b & 1) {
                res = Mono::op(res, a);
            }
            a = Mono::op(a, a);
            b >>= 1;
        }
        return res;
    }
}
```

2 模运算

2.1 Barrett 模乘

```
class Barrett {
public:
    constexpr Barrett(uint32_t Mod) : mod_(Mod), mu_(static_cast<uint64_t>(-1) / Mod + 1) {}

    constexpr uint32_t mod() const {
        return mod_;
    }

    constexpr uint32_t multiply(uint32_t a, uint32_t b) const {
        uint64_t z = static_cast<uint64_t>(a) * b;
        uint64_t q = static_cast<uint64_t>((static_cast<__uint128_t>(z) * mu_) >> 64);
        uint64_t y = q * mod_;
        return static_cast<uint32_t>(z - y + (z < y ? mod_ : 0));
    }

private:
    uint32_t mod_;
    uint64_t mu_;
};
```

2.2 运算

```
i64 mul(i64 a, i64 b, i64 m) {
    return static_cast<i128>(a) * b % m;
}

```

```
i64 power(i64 a, i64 b, i64 m) {
    i64 res = 1 % m;
    while (b > 0) {
        if (b & 1) {
            res = mul(res, a, m);
        }
        a = mul(a, a, m);
        b >>= 1;
    }
    return res;
}

template <typename T, std::integral U>
constexpr T power(T a, U b) {
    assert(b >= 0);
    T res{1};
    while (b) {
        if (b & 1) {
            res *= a;
        }
        b >>= 1;
        a *= a;
    }
    return res;
}
```

2.3 模数类

```
template <u32 P>
struct Mint {
    u32 v;
    Mint(i64 v_ = 0) : v((v_ %= P) < 0 ? v_ + P : v_) {}
    Mint inv() const {
        i64 x, y;
        exgcd(v, P, x, y);
        return x;
    }
    Mint& operator+=(Mint rhs) & {
        v += rhs.v;
        if (v >= P) v -= P;
        return *this;
    }
    Mint& operator-=(Mint rhs) & {
        v -= rhs.v;
        if (v >= P) v += P;
        return *this;
```

```
    }
    Mint& operator*=(Mint rhs) & {
        v = u64(v) * rhs.v % P;
        return *this;
    }
    Mint& operator/=(Mint rhs) & {
        return *this *= rhs.inv();
    }
    friend Mint operator-(Mint rhs) { return rhs.v ? Mint(P - rhs.v) : rhs; }
    friend Mint operator+(Mint lhs, Mint rhs) { return lhs += rhs; }
    friend Mint operator-(Mint lhs, Mint rhs) { return lhs -= rhs; }
    friend Mint operator*(Mint lhs, Mint rhs) { return lhs *= rhs; }
    friend Mint operator/(Mint lhs, Mint rhs) { return lhs /= rhs; }
    friend bool operator==(Mint lhs, Mint rhs) { return lhs.v == rhs.v; }
    friend bool operator!=(Mint lhs, Mint rhs) { return lhs.v != rhs.v; }
    friend istream& operator>>(istream& is, Mint& v) { i64 x; is >> x; v = x; return is; }
    friend ostream& operator<<(ostream& os, const Mint& v) { return os << v.v; }
};

constexpr u32 P = 998244353;
using Fp = Mint<P>;
```

2.4 all inverse

```
template <typename Fp>
vector<Fp> allInverse(const vector<Fp>& a) {
    for (auto& v : a) {
        assert(v != 0);
    }
    int n = a.size();
    vector<Fp> res(n + 1);
    res[0] = Fp{1};
    for (int i = 0; i < n; i++) {
        res[i + 1] = res[i] * a[i];
    }
    Fp inv = res[n].inverse();
    res.pop_back();
    for (int i = n - 1; i >= 0; i--) {
        res[i] *= inv;
        inv *= a[i];
    }
    return res;
}
```

3 图论

3.1 Graph 类

```
template <typename T>
struct Edge {
    int u, v;
    T w;
};

template <typename T = int, bool dir = true>
struct Graph {
    int n;
    vector<Edge<T>> edges;
    vector<vector<pair<int, int>>> adj;

    Graph() : n(0) {}
    Graph(int n_) : n(n_), adj(n_) {}

    int add(int u, int v, T w = {1}) {
        int id = edges.size();
        edges.emplace_back(u, v, w);
        adj[u].emplace_back(v, id);
        if constexpr (!dir) {
            adj[v].emplace_back(u, id);
        }
        return id;
    }

    auto& operator[](int u) const {
        return adj[u];
    }
};

template <typename T>
using Digraph = Graph<T, true>;
template <typename T>
using Undigraph = Graph<T, false>;

template <typename T>
Digraph<T> reverse(const Digraph<T>& g) {
    Digraph<T> rg(g.n);
    for (auto& [u, v, w] : g.edges) {
        rg.add(v, u, w);
    }
    return rg;
```

```cpp
}

template <typename T>
vector<int> degree(const Undigraph<T>& g) {
    vector<int> deg(g.n);
    for (auto& e : g.edges) {
        deg[e.u]++;
        deg[e.v]++;
    }
    return deg;
}

template <typename T>
pair<vector<int>, vector<int>> degree(const Digraph<T>& g) {
    vector<int> in(g.n), out(g.n);
    for (auto& e : g.edges) {
        in[e.v]++;
        out[e.u]++;
    }
    return {in, out};
}

template <typename T, bool dir>
Graph<T, dir> induced(const Graph<T, dir>& g, const vector<int>& a) {
    int n = a.size();
    vector<int> id(g.n, -1);
    for (int i = 0; i < n; i++) {
        id[a[i]] = i;
    }
    Graph<T, dir> res(n);
    for (auto& e : g.edges) {
        if (id[e.u] != -1 && id[e.v] != -1) {
            res.add(id[e.u], id[e.v], e.w);
        }
    }
    return res;
}

template <typename T>
Undigraph<T> toUndigraph(const Digraph<T>& g) {
    Undigraph<T> res(g.n);
    for (auto& e : g.edges) {
        res.add(e.u, e.v, e.w);
    }
    return res;
}
```

```cpp
}
```

3.2 线段树优化建图

```cpp
template <typename T>
struct RangeGraph {
    int n, tot;
    vector<Edge<T>> edges;

    RangeGraph() {}
    RangeGraph(int n_) : n(n_), tot(3 * n) {
        for (int i = 2; i < 2 * n; i++) {
            add(out(i / 2), out(i), T{});
            add(in(i), in(i / 2), T{});
        }
    }

    int out(int i) {
        return (i >= n ? i - n : n + i);
    }
    int in(int i) {
        return (i >= n ? i - n : 2 * n + i);
    }

    int add(int u, int v, T w = {1}) {
        int id = edges.size();
        edges.emplace_back(u, v, w);
        return id;
    }

    void toRange(int u, int l, int r, T w = {1}) {
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) {
                add(u, out(l++), w);
            }
            if (r & 1) {
                add(u, out(--r), w);
            }
        }
    }

    void fromRange(int l, int r, int v, T w = {1}) {
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) {
                add(in(l++), v, w);
            }
```

```
            if (r & 1) {
                add(in(--r), v, w);
            }
        }
    }

    void RangeToRange(int l1, int r1, int l2, int r2, T w = {1}) {
        int u = tot++;
        fromRange(l1, r1, u, w);
        toRange(u, l2, r2, T{});
    }

    Digraph<T> build() {
        Digraph<T> g(tot);
        for (auto& [u, v, w] : edges) {
            g.add(u, v, w);
        }
        return g;
    }
};
```

3.3 最短路

3.3.1 Dijkstra

```
template <typename T, bool dir>
pair<vector<T>, vector<int>> dijkstra(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    using P =  pair<T, int>;
    priority_queue<P, vector<P>, greater<P>> pq;

    dist[s] = 0;
    pq.emplace(0, s);

    while (!pq.empty()) {
        auto [d, u] = pq.top();
        pq.pop();
        if (d != dist[u]) {
            continue;
        }
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                pq.emplace(dist[v], v);
            }
        }
    }

    return {dist, pre};
}

template <typename T, bool dir>
pair<vector<T>, vector<int>> dijkstraDense(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    vector<int> done(n, false);

    dist[s] = 0;

    for (int _ = 0; _ < n; _++) {
        int u = -1;
        for (int v = 0; v < n; v++) {
            if (!done[v] && (u == -1 || dist[v] < dist[u])) {
                u = v;
            }
        }
        if (u == -1 || dist[u] == inf<T>) {
            break;
        }
        done[u] = true;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
            }
        }
    }

    return {dist, pre};
}

template <typename T, bool dir>
tuple<vector<T>, vector<int>, vector<int>> dijkstra(const Graph<T, dir>& g,
vector<int>& s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
```

```
    vector<int> pre(n, -1), root(n, -1);
    using P =  pair<T, int>;
    priority_queue<P, vector<P>, greater<P>> pq;

    for (auto& u : s) {
        dist[u] = 0;
        root[u] = u;
        pq.emplace(0, u);
    }

    while (!pq.empty()) {
        auto [d, u] = pq.top();
        pq.pop();
        if (d != dist[u]) {
            continue;
        }
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                root[v] = root[u];
                pq.emplace(dist[v], v);
            }
        }
    }

    return {dist, pre, root};
}
```

3.3.2 01-BFS

```
template <typename T, bool dir>
pair<vector<T>, vector<int>> bfs_01(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);
    deque<int> que;

    dist[s] = 0;
    que.push_front(s);

    while (!que.empty()) {
        auto u = que.front();
        que.pop_front();
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                if (w == 0) {
                    que.push_front(v);
                } else {
                    que.push_back(v);
                }
            }
        }
    }

    return {dist, pre};
}

template <typename T, bool dir>
tuple<vector<T>, vector<int>, vector<int>> bfs_01(const Graph<T, dir>& g,
vector<int>& s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1), root(n, -1);
    deque<int> que;

    for (auto& u : s) {
        dist[u] = 0;
        root[u] = u;
        que.push_front(u);
    }

    while (!que.empty()) {
        auto u = que.front();
        que.pop_front();
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                root[v] = root[u];
                if (w == 0) {
                    que.push_front(v);
                } else {
                    que.push_back(v);
                }
            }
        }
```

```
        }
    }

    return {dist, pre, root};
}
```

3.3.3 Bellman-Ford

```
template <typename T, bool dir>
pair<vector<T>, vector<int>> bellmanFord(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);

    dist[s] = 0;

    for (int round = 1;; round++) {
        bool upd = false;
        for (int u = 0; u < n; u++) {
            if (dist[u] == inf<T>) {
                continue;
            }
            for (auto [v, id] : g[u]) {
                T w = g.edges[id].w;
                if (dist[v] > dist[u] + w) {
                    dist[v] = dist[u] + w;
                    pre[v] = u;
                    upd = true;
                    if (round >= n) {
                        return {{}, {}};
                    }
                }
            }
        }
        if (!upd) {
            break;
        }
    }

    return {dist, pre};
}
```

3.3.4 SPFA

```
template <typename T, bool dir>
pair<vector<T>, vector<int>> spfa(const Graph<T, dir>& g, int s) {
    int n = g.n;
    vector<T> dist(n, inf<T>);
    vector<int> pre(n, -1);

    dist[s] = 0;

    queue<int> q;
    vector<char> inq(n, false);
    vector<int> cnt(n, 0);
    q.push(s);
    inq[s] = true;

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        inq[u] = false;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pre[v] = u;
                if (!inq[v]) {
                    q.push(v);
                    inq[v] = true;
                    cnt[v]++;
                    if (cnt[v] >= n) {
                        return {{}, {}};
                    }
                }
            }
        }
    }

    return {dist, pre};
}
```

3.3.5 Floyd

```
template <typename T>
void floyd(vector<vector<T>>& dist) {
    int n = dist.size();
    for (int k = 0; k < n; k++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (dist[i][k] < inf<T> && dist[k][j] < inf<T>) {
                    dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
                }
```

```
            }
        }
    }
}

template <typename T, bool dir>
vector<vector<T>> floyd(const Graph<T, dir>& g) {
    int n = g.n;
    vector dist(n, vector<T>(n, inf<T>));
    for (int u = 0; u < n; u++) {
        dist[u][u] = 0;
        for (auto [v, id] : g[u]) {
            T w = g.edges[id].w;
            dist[u][v] = min(dist[u][v], w);
        }
    }
    floyd(dist);
    return dist;
}

```

3.3.6 路径重建

```
vector<int> restorePath(vector<int>& pre, int t) {
    vector<int> path = {t};
    while (pre[path.back()] != -1) {
        path.push_back(pre[path.back()]);
    }
    reverse(path.begin(), path.end());
    return path;
}
```

3.4 差分约束

```
template <typename T>
struct DifferenceConstraints {
    int n, s;
    Digraph<T> graph;

    DifferenceConstraints() {}
    DifferenceConstraints(int n_) : n(n_), s(n), graph(n + 1) {
        for (int u = 0; u < n; u++) {
            graph.add(s, u);
        }
    }

    // x_v - x_u <= w
    void constraint(int u, int v, T w) {
        graph.add(u, v, w);
    }

    // x_v - x_u >= w
    void lowerBound(int u, int v, T w) {
        graph.add(v, u, -w);
    }

    // x_v - x_u = w
    void equality(int u, int v, T w) {
        constraint(u, v, w);
        lowerBound(u, v, w);
    }

    pair<bool, vector<T>> solve() {
        auto [dist, pre] = bellmanFord<T>(graph, s);
        if (dist.empty()) {
            return {false, {}};
        }
        dist.pop_back();
        return {true, dist};
    }
};
```

3.5 拓扑排序

```
template <typename T>
vector<int> toposort(const Digraph<T>& g) {
    int n = g.n;
    vector<int> topo, deg(n, 0);
    for (int u = 0; u < n; u++) {
        for (auto [v, id] : g[u]) {
            deg[v]++;
        }
    }
    for (int u = 0; u < n; u++) {
        if (deg[u] == 0) {
            topo.push_back(u);
        }
    }
    for (int i = 0; i < int(topo.size()); i++) {
        int u = topo[i];
        for (auto [v, id] : g[u]) {
            if (--deg[v] == 0) {
                topo.push_back(v);
```

```
            }
        }
    }
    return (int(topo.size()) != n ? vector<int>() : topo);
}

template <typename T>
vector<int> lexMinToposort(const Digraph<T>& g) {
    int n = g.n;
    vector<int> topo, deg(n, 0);
    for (int u = 0; u < n; u++) {
        for (auto [v, id] : g[u]) {
            deg[v]++;
        }
    }
    priority_queue<int, vector<int>, greater<int>> pq;
    for (int u = 0; u < n; u++) {
        if (deg[u] == 0) {
            pq.push(u);
        }
    }
    while (!pq.empty()) {
        int u = pq.top();
        pq.pop();
        topo.push_back(u);
        for (auto [v, id] : g[u]) {
            if (--deg[v] == 0) {
                pq.push(v);
            }
        }
    }
    return (int(topo.size()) != n ? vector<int>() : topo);
}
```

3.6 欧拉通路/回路

给定起点 s，还需要满足所有有边的顶点属于同一个连通分量，且 s 不能是孤立点。

无向图：

- 奇度点的数量一定是偶数，因此不会出现只有 1 个奇度点的情况。
- 有 0 个奇度点时，存在从任意有边顶点 s 出发的欧拉回路。
- 有 2 个奇度点时，s 必须是其中一个，存在从 s 出发的欧拉通路。
- 奇度点超过 2 个时不存在欧拉通路。

有向图：

- 若所有顶点满足 $\text{in} = \text{out}$，存在从任意有边顶点 s 出发的欧拉回路。
- 若恰有一个顶点满足 $\text{out} = \text{in} + 1$，它必须是 s；同时恰有一个顶点满足 $\text{in} = \text{out} + 1$，存在从 s 出发的欧拉通路。
- 其他情况不存在欧拉通路。

```
// 0: no Eulerian walk, 1: Eulerian path, 2: Eulerian circuit
template <typename T, bool dir>
int isEulerian(const Graph<T, dir>& g, int s = -1) {
    int n = g.n;
    if (g.edges.empty()) {
        return 2;  // 无边：任意点都有一条长度 0 的平凡欧拉回路
    }
    vector<int> deg(n);
    DSU dsu(n);
    for (auto& e : g.edges) {
        dsu.merge(e.u, e.v);
        deg[e.u]++;
        deg[e.v] += (dir ? -1 : 1);
    }
    int root = g.edges[0].u;
    for (auto& e : g.edges) {
        if (!dsu.same(root, e.u)) {
            return 0;
        }
    }
    if (s != -1 && !dsu.same(root, s)) {
        return 0;
    }
    int diff = 0;
    for (int u = 0; u < n; u++) {
        diff += dir ? abs(deg[u]) : (deg[u] & 1);
    }
    if (diff == 0) {
        return 2;
    }
    if (diff != 2) {
        return 0;
    }
    if (s == -1) {
        return 1;
    }
    return (dir ? deg[s] == 1 : (deg[s] & 1));
}

template <typename T, bool dir>
bool isEulerianPath(const Graph<T, dir>& g, int s = -1) {
```

```
    return isEulerian(g, s) >= 1;
}
template <typename T, bool dir>
bool isEulerianCircuit(const Graph<T, dir>& g, int s = -1) {
    return isEulerian(g, s) == 2;
}

template <typename T, bool dir>
pair<vector<int>, vector<int>> findEulerian(const Graph<T, dir>& g, int s = -1) {
    int n = g.n;
    int m = g.edges.size();
    assert(n > 0);
    assert(s == -1 || (0 <= s && s < n));
    if (s == -1) {
        vector<int> deg(n);
        for (auto& e : g.edges) {
            deg[e.u]++;
            deg[e.v] += (dir ? -1 : 1);
        }
        if constexpr (dir) {
            s = max_element(deg.begin(), deg.end()) - deg.begin();
            if (deg[s] == 0) {
                s = m == 0 ? 0 : g.edges[0].u;
            }
        } else {
            s = (m == 0 ? 0 : g.edges[0].u);
            for (int u = 0; u < n; u++) {
                if (deg[u] & 1) {
                    s = u;
                    break;
                }
            }
        }
    }
    if (m == 0) {
        return {{s}, {}};
    }
    vector<int> bal(n), ptr(n), vs, es;
    vector<char> used(m, false);
    vector<pair<int, int>> stk = {{s, -1}};
    bal[s]++;
    while (!stk.empty()) {
        auto [u, eid] = stk.back();
        if (ptr[u] == int(g.adj[u].size())) {
            vs.push_back(u);
            if (eid != -1) {
                es.push_back(eid);
            }
            stk.pop_back();
        } else {
            auto [v, id] = g.adj[u][ptr[u]++];
            if (used[id]) {
                continue;
            }
            bal[u]--;
            bal[v]++;
            used[id] = true;
            stk.emplace_back(v, id);
        }
    }
    for (auto& x : bal) {
        if (x < 0) {
            return {{}, {}};
        }
    }
    if (int(vs.size()) != m + 1) {
        return {{}, {}};
    }
    reverse(vs.begin(), vs.end());
    reverse(es.begin(), es.end());
    return {vs, es};
}
```

3.7 最小生成树

3.7.1 Kruskal

```
template <typename T>
pair<vector<int>, T> kruskal(const Undigraph<T>& g) {
    vector<int> ord(g.edges.size());
    iota(ord.begin(), ord.end(), 0);
    sort(ord.begin(), ord.end(), [&](int i, int j) {
        return g.edges[i].w < g.edges[j].w;
    });

    T cost = 0;
    vector<int> res;
    DSU dsu(g.n);
    for (auto& id : ord) {
        auto& e = g.edges[id];
        if (dsu.merge(e.u, e.v)) {
```

```
            res.push_back(id);
            cost += e.w;
        }
    }
    return {res, cost};
}
```

3.7.2 Prim

```
template <typename T, typename F>
vector<Edge<T>> prim(int n, F&& cost) {
    vector<Edge<T>> res;
    vector<T> wei(n, inf<T>);
    vector<int> pre(n);

    auto add = [&](int u) -> void {
        for (int v = 0; v < n; v++) {
            if (pre[v] == -1) {
                continue;
            }
            T w = cost(u, v);
            if (wei[v] > w) {
                wei[v] = w;
                pre[v] = u;
            }
        }
        wei[u] = inf<T>;
        pre[u] = -1;
    };

    add(0);
    for (int i = 0; i < n - 1; i++) {
        int u = min_element(wei.begin(), wei.end()) - wei.begin();
        res.emplace_back(pre[u], u, wei[u]);
        add(u);
    }

    return res;
}

template <typename T>
pair<vector<int>, T> prim(Undigraph<T>& g) {
    int n = g.n;
    vector<int> res, pre(n, -2);
    vector<T> wei(n, inf<T>);
    T cost = 0;
    auto add = [&](int u) -> void {
        for (auto& [v, id] : g[u]) {
            if (pre[v] == -1) {
                continue;
            }
            T w = g.edges[id].w;
            if (wei[v] > w) {
                wei[v] = w;
                pre[v] = id;
            }
        }
        wei[u] = inf<T>;
        pre[u] = -1;
    };
    add(0);
    for (int i = 0; i < n - 1; i++) {
        int u = min_element(wei.begin(), wei.end()) - wei.begin();
        if (wei[u] == inf<T>) {
            break;
        }
        res.push_back(pre[u]);
        cost += wei[u];
        add(u);
    }
    return {res, cost};
}
```

3.7.3 Borůvka

```
template <typename T>
pair<vector<int>, T> boruvka(Undigraph<T>& g) {
    auto& edges = g.edges;
    int n = g.n;
    T cost = 0;
    vector<int> res;
    DSU dsu(n);
    int cnt = n;

    while (cnt > 1) {
        vector<int> best(n, -1);
        int m = edges.size();

        for (int i = 0; i < m; i++) {
            auto& e = edges[i];
            int u = dsu.find(e.u);
            int v = dsu.find(e.v);
```

```
            if (u == v) {
                continue;
            }
            if (best[u] == -1 || e.w < edges[best[u]].w) {
                best[u] = i;
            }
            if (best[v] == -1 || e.w < edges[best[v]].w) {
                best[v] = i;
            }
        }

        bool upd = false;
        for (int u = 0; u < n; u++) {
            if (best[u] == -1) {
                continue;
            }
            auto& e = edges[best[u]];
            if (dsu.merge(e.u, e.v)) {
                res.push_back(best[u]);
                cost += e.w;
                cnt--;
                upd = true;
            }
        }
        if (!upd) {
            break;
        }
    }

    return {res, cost};
}
```

3.7.4 Kruskal 重构树

```
template <typename T>
pair<Undigraph<int>, vector<T>> kruskalTree(const Undigraph<T>& g) {
    int n = g.n;
    auto [ids, cost] = kruskal(g);
    int c = ids.size();

    Undigraph<int> res(n + c);
    vector<T> val(n + c);
    DSU dsu(n);
    vector<int> top(n);
    iota(top.begin(), top.end(), 0);

    for (int i = 0; i < c; i++) {
        int id = ids[i];
        int u = g.edges[id].u, v = g.edges[id].v;
        u = dsu.find(u), v = dsu.find(v);
        int p = n + i;
        res.add(p, top[u]);
        if (u != v) {
            res.add(p, top[v]);
        }
        dsu.merge(u, v);
        top[dsu.find(u)] = p;
        val[p] = g.edges[id].w;
    }

    return {res, val};
}
```

3.8 连通性

3.8.1 强连通分量

```
template <typename T>
pair<int, vector<int>> SCC(const Digraph<T>& g) {
    int n = g.n;
    vector<int> stk, dfn(n, -1), low(n), comp(n, -1);
    int cur = 0, cnt = 0;
    auto dfs = [&](auto&& self, int u) -> void {
        dfn[u] = low[u] = cur++;
        stk.push_back(u);
        for (auto& [v, id] : g[u]) {
            if (dfn[v] == -1) {
                self(self, v);
                low[u] = min(low[u], low[v]);
            } else if (comp[v] == -1) {
                low[u] = min(low[u], dfn[v]);
            }
        }
        if (dfn[u] == low[u]) {
            int y;
            do {
                y = stk.back();
                comp[y] = cnt;
                stk.pop_back();
            } while (y != u);
            cnt++;
        }
```

```
    };
    for (int i = 0; i < n; i++) {
        if (dfn[i] == -1) {
            dfs(dfs, i);
        }
    }
    for (auto& u : comp) {
        u = cnt - 1 - u;
    }
    return {cnt, comp};
}
```

3.8.2 边双连通分量

```
template <typename T>
pair<int, vector<int>> EBCC(const Undigraph<T>& g) {
    int n = g.n;
    vector<int> stk;
    vector<int> dfn(n, -1), low(n), comp(n, -1);
    int cur = 0, cnt = 0;
    auto dfs = [&](auto&& self, int u, int parentEdge) -> void {
        dfn[u] = low[u] = cur++;
        stk.push_back(u);
        for (auto [v, id] : g[u]) {
            if (id == parentEdge) {
                continue;
            }
            if (dfn[v] == -1) {
                self(self, v, id);
                low[u] = min(low[u], low[v]);
            } else {
                low[u] = min(low[u], dfn[v]);
            }
        }
        if (dfn[u] == low[u]) {
            int v;
            do {
                v = stk.back();
                stk.pop_back();
                comp[v] = cnt;
            } while (v != u);
            cnt++;
        }
    };
    for (int u = 0; u < n; u++) {
        if (dfn[u] == -1) {
            dfs(dfs, u, -1);
        }
    }
    return {cnt, comp};
}
```

3.8.3 圆方树

```
struct BlockCutTree {
    int n, tot;
    Undigraph<int> tree;

    BlockCutTree() {}
    template <typename T>
    BlockCutTree(const Undigraph<T>& g) : n(g.n), tree(n) {
        vector<int> dfn(n, -1), low(n), stk;
        int cur = 0;
        auto dfs = [&](auto&& self, int u) -> void {
            dfn[u] = low[u] = cur++;
            stk.push_back(u);
            for (auto [v, id] : g[u]) {
                if (dfn[v] == -1) {
                    self(self, v);
                    low[u] = min(low[u], low[v]);
                    if (low[v] >= dfn[u]) {
                        int b = tree.n++;
                        tree.adj.emplace_back();
                        tree.add(u, b);
                        int x;
                        do {
                            x = stk.back();
                            stk.pop_back();
                            tree.add(x, b);
                        } while (x != v);
                    }
                } else {
                    low[u] = min(low[u], dfn[v]);
                }
            }
        };
        for (int u = 0; u < n; u++) {
            if (dfn[u] == -1) {
                dfs(dfs, u);
                stk.pop_back();
            }
```

```
        }
        tot = tree.n;
    }

    bool isCut(int u) const {
        return tree[u].size() > 1;
    }
};
```

3.9 2-SAT

`addClause(u, f, v, g)` 加入一个析取子句：

$$(x_u = f) \vee (x_v = g).$$

这里 `x_i` 是第 `i` 个布尔变量，变量编号从 0 开始。`f` 和 `g` 决定文字的正负： `true` 表示正文字 x_i，`false` 表示负文字 $\neg x_i$。

例如： `addClause(3, true, 5, false)` 表示

$$x_3 \vee \neg x_5.$$

一个子句 $(A \vee B)$ 等价于两条蕴含边：

$$(\neg A) \to B, \quad (\neg B) \to A.$$

所以 `addClause(u, f, v, g)` 实际加入：

$$(x_u = \neg f) \to (x_v = g), \quad (x_v = \neg g) \to (x_u = f).$$

图中的编号约定是：$2i$ 表示变量 x_i 取假值，$2i+1$ 表示变量 x_i 取真值。 因此第一条边的起点是第一个文字的否定，终点是第二个文字本身；另一条边同理。

```
struct TwoSat {
    int n;
    Digraph<int> graph;
    vector<int> ans;

    TwoSat(int n_) : n(n_), graph(2 * n), ans(n) {}

    // add clause: (x_u = f) ∨ (x_v = g)
    void addClause(int u, bool f, int v, bool g) {
        graph.add(2 * u + !f, 2 * v + g);
        graph.add(2 * v + !g, 2 * u + f);
    }

    // (x_u = f)
    void set(int u, bool f) {
        addClause(u, f, u, f);
    }

    // (x_u = f) -> (x_v = g)
    void implies(int u, int f, int v, int g) {
        addClause(u, !f, v, g);
    }

    pair<bool, vector<int>> satisfiable() {
        auto [cnt, id] = SCC(graph);
        for (int i = 0; i < n; i++) {
            if (id[2 * i] == id[2 * i + 1]) {
                return {false, {}};
            }
            ans[i] = id[2 * i] < id[2 * i + 1];
        }
        return {true, ans};
    }
};
```

3.10 树上问题

3.10.1 Tree 类

```
template <typename T>
struct Tree {
    Undigraph<T>& g;
    int n, root;
    vector<int> fa, pe, dep, siz, in, out, seq;
    vector<T> dist;

    Tree(Undigraph<T>& g, int rt = 0) : g(g) {
        build(rt);
    }

    void build(int rt = 0) {
        n = g.n;
        root = rt;
        assert(0 <= root && root < n);
        assert(int(g.edges.size()) == n - 1);
        fa.assign(n, -1);
        pe.assign(n, -1);
        dep.assign(n, 0);
        siz.assign(n, 1);
        in.resize(n);
        out.resize(n);
        seq.resize(n);
        dist.assign(n, T{});

        int cur = 0;
        auto dfs = [&](auto&& self, int u) -> void {
            in[u] = cur;
            seq[cur++] = u;
            for (auto [v, id] : g[u]) {
```

```
                if (v == fa[u]) {
                    continue;
                }
                fa[v] = u;
                pe[v] = id;
                dep[v] = dep[u] + 1;
                dist[v] = dist[u] + g.edges[id].w;
                self(self, v);
                siz[u] += siz[v];
            }
            out[u] = cur;
        };
        dfs(dfs, root);
    }

    int eToV(int i) {
        auto& e = g.edges[i];
        return (dep[e.u] > dep[e.v] ? e.u : e.v);
    }

    int vToE(int u) {
        return pe[u];
    }

    vector<int> subtree(int u) {
        return vector<int>(seq.begin() + in[u], seq.begin() + out[u]);
    }
};
```

3.10.2 树链剖分

```
template <typename T>
struct HLD {
    Tree<T>& tree;
    vector<int> he, top;

    HLD(Tree<T>& tree) : tree(tree) {
        build();
    }

    void build() {
        int n = tree.n;
        he.assign(n, -1);
        top.resize(n);
        for (int v = 0; v < n; v++) {
            int u = tree.fa[v];
            if (u == -1) {
                continue;
            }
            if (he[u] == -1 || tree.siz[v] > tree.siz[he[u]]) {
                he[u] = v;
            }
        }
        int cur = 0;
        auto dfs = [&](auto&& self, int u, int t) -> void {
            top[u] = t;
            tree.in[u] = cur;
            tree.seq[cur++] = u;
            if (he[u] != -1) {
                self(self, he[u], t);
            }
            for (auto [v, id] : tree.g[u]) {
                if (v == tree.fa[u] || v == he[u]) {
                    continue;
                }
                self(self, v, v);
            }
            tree.out[u] = cur;
        };
        dfs(dfs, tree.root, tree.root);
    }

    int lca(int u, int v) {
        while (top[u] != top[v]) {
            if (tree.dep[top[u]] < tree.dep[top[v]]) {
                swap(u, v);
            }
            u = tree.fa[top[u]];
        }
        return tree.dep[u] < tree.dep[v] ? u : v;
    }

    int dist(int u, int v) {
        return tree.dep[u] + tree.dep[v] - 2 * tree.dep[lca(u, v)];
    }

    int jump(int u, int k) {
        if (k > tree.dep[u]) {
            return -1;
        }
```

```
        int d = tree.dep[u] - k;
        while (tree.dep[top[u]] > d) {
            u = tree.fa[top[u]];
        }
        return tree.seq[tree.in[u] - tree.dep[u] + d];
    }

    int jump(int u, int v, int k) {
        int p = lca(u, v);
        int a = tree.dep[u] - tree.dep[p];
        int b = tree.dep[v] - tree.dep[p];
        if (k > a + b) {
            return -1;
        }
        if (k <= a) {
            return jump(u, k);
        }
        return jump(v, a + b - k);
    }

    bool isAncester(int u, int v) {
        return tree.in[u] <= tree.in[v] && tree.in[v] < tree.out[u];
    }

    int rootedParent(int root, int v) {
        if (root == v) {
            return v;
        }
        if (!isAncester(v, root)) {
            return tree.fa[v];
        }
        return jump(root, tree.dep[root] - tree.dep[v] - 1);
    }

    int rootedSize(int root, int v) {
        if (root == v) {
            return tree.n;
        }
        if (!isAncester(v, root)) {
            return tree.siz[v];
        }
        return tree.n - tree.siz[rootedParent(root, v)];
    }

    int rootedLca(int root, int u, int v) {
        return lca(root, u) ^ lca(root, v) ^ lca(u, v);
    }

    template <typename F>
    void forPath(int u, int v, bool edge, F&& f) {
        vector<pair<int, int>> down;
        while (top[u] != top[v]) {
            if (tree.dep[top[u]] > tree.dep[top[v]]) {
                f(tree.in[top[u]], tree.in[u] + 1, true);
                u = tree.fa[top[u]];
            } else {
                down.emplace_back(tree.in[top[v]], tree.in[v] + 1);
                v = tree.fa[top[v]];
            }
        }
        bool rev = tree.dep[u] > tree.dep[v];
        int l = min(tree.in[u], tree.in[v]) + edge;
        int r = max(tree.in[u], tree.in[v]) + 1;
        if (l < r) {
            f(l, r, rev);
        }
        for (int i = int(down.size()) - 1; i >= 0; i--) {
            auto [l, r] = down[i];
            f(l, r, false);
        }
    }
};
```

3.10.3 树上启发式合并

适用于静态树上的离线子树统计，如不同颜色数、颜色频次、众数等。保留重子树的数据，暴力补入轻子树的数据。

- `add(u)`：加入节点 `u` 的数据，不应依赖加入顺序。
- `query(u)`：当前数据恰好包含 `u` 的整棵子树，每个节点一次；据此记录答案，不破坏统计状态。
- `reset()`：清空全部统计状态，建议记录修改过的位置，避免每次清空整个值域。

调用前统计状态须为空，结束后也为空。`add` 共调用 $O(n \log n)$ 次，`query` 调用 n 次；若单次加入、查询为 $O(1)$，且清空成本可摊到加入操作上，总时间为 $O(n \log n)$。

```
template <typename T, typename F1, typename F2, typename F3>
void dsuOnTree(const HLD<T>& hld, F1&& add, F2&& query, F3&& reset) {
    const auto& tree = hld.tree;
    for (int i = tree.n - 1; i >= 0; i--) {
        int u = tree.seq[i];
        add(u);
        for (auto& [v, id] : tree.g[u]) {
```

```
            if (v == tree.fa[u] || v == hld.he[u]) {
                continue;
            }
            for (int j = tree.in[v]; j < tree.out[v]; j++) {
                add(tree.seq[j]);
            }
        }
        query(u);
        if (hld.top[u] == u) {
            reset();
        }
    }
}
```

3.10.4 点分治

回调 `f(fa, vt, pos)`：`vt[i]` 为原树编号，`vt[0]` 为重心；`fa[i]` 为局部父亲编号，`fa[0] = -1`，其余满足 `0 <= fa[i] < i`。从 $b = 1$ 起，`[pos[b], pos[b+1])` 为各分支。只统计经过重心的路径，排除同分支点对；不应修改数组或保留引用。

当前块大小为 m，单次回调 $O(m)$ 时总时间 $O(n \log n)$，$O(m \log m)$ 时为 $O(n \log^2 n)$。初始化、清空也计入回调开销，避免每次扫描全局数组。

```
// f(fa, vt, pos)，局部编号 0 为重心，各分支占连续区间。
template <typename T, typename F>
void centroidDecompsition(const Tree<T>& tree, F&& f) {
    const auto& g = tree.g;
    vector<int> dead(tree.n), siz, fa, vt, pos;
    auto collect = [&](int u, int p) {
        int l = vt.size();
        vt.push_back(u);
        fa.push_back(p);
        for (int i = l; i < int(vt.size()); i++) {
            int u = vt[i];
            for (auto [v, id] : g[u]) {
                if (dead[v] || (fa[i] != -1 && v == vt[fa[i]])) {
                    continue;
                }
                vt.push_back(v);
                fa.push_back(i);
            }
        }
    };
    vector<int> rts{tree.root};
    for (int t = 0; t < int(rts.size()); t++) {
        vt.clear();
        fa.clear();
        collect(rts[t], -1);
        int m = vt.size(), c = -1;
        siz.assign(m, 1);
        for (int i = m - 1; i >= 0; i--) {
            if (siz[i] >= (m + 1) / 2) {
                c = vt[i];
                break;
            }
            siz[fa[i]] += siz[i];
        }
        dead[c] = true;
        vt = {c};
        fa = {-1};
        pos = {0, 1};
        for (auto [v, id] : g[c]) {
            if (!dead[v]) {
                rts.push_back(v);
                collect(v, 0);
                pos.push_back(vt.size());
            }
        }
        f(fa, vt, pos);
    }
}
```

3.10.5 LCA

```
template <typename T>
struct LCA {
    Tree<T>& tree;
    int n, logn;
    vector<vector<int>> fa;

    LCA(Tree<T>& t) : tree(t), n(tree.n), logn(__lg(n)), fa(n, vector<int>(logn + 1)) {
        for (int u = 0; u < n; u++) {
            fa[u][0] = (tree.fa[u] == -1 ? u : tree.fa[u]);
        }
        for (int k = 1; k <= logn; k++) {
            for (int u = 0; u < n; u++) {
                fa[u][k] = fa[fa[u][k - 1]][k - 1];
            }
        }
    }

    int get(int u, int v) {
        if (tree.dep[u] < tree.dep[v]) {
```

```
            swap(u, v);
        }
        for (int i = logn; i >= 0; i--) {
            if (tree.dep[fa[u][i]] >= tree.dep[v]) {
                u = fa[u][i];
            }
        }
        if (u == v) {
            return u;
        }
        for (int i = logn; i >= 0; i--) {
            if (fa[u][i] != fa[v][i]) {
                u = fa[u][i];
                v = fa[v][i];
            }
        }
        return fa[u][0];
    }

    int dist(int u, int v) {
        int res = tree.dep[u] + tree.dep[v] - 2 * tree.dep[get(u, v)];
        return res;
    }
};

template <typename T>
struct FastLCA {
    struct DfnMin {
        struct Type {
            int dfn, fa;
        };
        static Type op(const Type& a, const Type& b) {
            return a.dfn < b.dfn ? a : b;
        }
    };

    Tree<T>& tree;
    int n;
    SparseTable<DfnMin> st;

    FastLCA(Tree<T>& tree) : tree(tree), n(tree.n) {
        vector<typename DfnMin::Type> seq(n);
        for (int u = 0; u < n; u++) {
            int p = tree.fa[u] == -1 ? u : tree.fa[u];
            seq[tree.in[u]] = {tree.in[p], p};
        }
        st = SparseTable<DfnMin>(seq);
    }

    int query(int u, int v) {
        if (u == v) {
            return u;
        }
        if (tree.in[u] > tree.in[v]) {
            swap(u, v);
        }
        return st.query(tree.in[u] + 1, tree.in[v] + 1).fa;
    }

    int dist(int u, int v) {
        return tree.dep[u] + tree.dep[v] - 2 * tree.dep[query(u, v)];
    }
};
```

3.10.6 树上倍增

```
template <typename Mono, typename T>
struct TreeLifting {
    using X = typename Mono::Type;

    Tree<T>& tree;
    BinaryLifting<Mono> lift;

    TreeLifting(Tree<T>& tree) : tree(tree), lift(tree.n) {
        int n = tree.n;
        for (int v = 0; v < n; v++) {
            if (v == tree.root) {
                lift.add(v, -1, Mono::e());
            } else {
                lift.add(v, tree.fa[v], Mono::from(tree.g.edges[tree.vToE(v)].w));
            }
        }
        lift.build();
    }

    int lca(int u, int v) {
        if (tree.dep[u] < tree.dep[v]) {
            swap(u, v);
        }
        u = lift.jump(u, tree.dep[u] - tree.dep[v]);
```

```
        if (u == v) {
            return u;
        }
        for (int k = lift.logn; k >= 0; k--) {
            if (lift.nxt[k][u] != lift.nxt[k][v]) {
                u = lift.nxt[k][u];
                v = lift.nxt[k][v];
            }
        }
        return lift.nxt[0][u];
    }

    pair<int, X> query(int u, int v) {
        int l = lca(u, v);
        X res = Mono::op(lift.query(u, tree.dep[u] - tree.dep[l]).second,
lift.query(v, tree.dep[v] - tree.dep[l]).second);
        return {l, res};
    }
};
```

3.10.7 虚树

```
template <typename T>
struct VirtualTree {
    TreeLifting<AddMono<i64>, T>& tree;
    vector<int> vt;
    vector<vector<pair<int, T>>> adj;

    VirtualTree(TreeLifting<AddMono<i64>, T>& t) : tree(t) {}

    pair<vector<int>&, vector<vector<pair<int, T>>>&> build(vector<int>& h) {
        vt = h;
        auto cmp = [&](int u, int v) { return tree.tree.in[u] < tree.tree.in[v]; };
        sort(vt.begin(), vt.end(), cmp);
        int m = vt.size();
        for (int i = 1; i < m; i++) {
            vt.push_back(tree.lca(vt[i - 1], vt[i]));
        }
        sort(vt.begin(), vt.end(), cmp);
        vt.erase(unique(vt.begin(), vt.end()), vt.end());
        int n = vt.size();
        adj.resize(n);
        for (auto& e : adj) {
            e.clear();
        }
        for (int i = 0; i < n - 1; i++) {
            int p = tree.lca(vt[i], vt[i + 1]);
            int u = lower_bound(vt.begin(), vt.end(), p, cmp) - vt.begin();
            int v = i + 1;
            T w = tree.tree.dep[vt[v]] - tree.tree.dep[p];
            adj[u].emplace_back(v, w);
            adj[v].emplace_back(u, w);
        }
        return {vt, adj};
    }
};
```

3.10.8 树上阿贝尔群

```
template <typename T, typename Abel, bool edge = false>
struct TreeAbel {
    using X = typename Abel::Type;

    HLD<T>& hld;
    Tree<T>& tree;
    int n;
    Fenwick<Abel> sub;
    DualFenwick<Abel> path;

    TreeAbel(HLD<T>& hld_) : hld(hld_), tree(hld.tree), n(tree.n), sub(n), path(n)
{}
    TreeAbel(HLD<T>& hld_, const vector<X>& init) : hld(hld_), tree(hld.tree),
n(tree.n) {
        assert(int(init.size()) == n - edge);
        vector<X> a(n, Abel::e()), b(n, Abel::e());
        for (int i = 0; i < int(init.size()); i++) {
            int u = (edge ? tree.eToV(i) : i);
            a[tree.in[u]] = init[i];
        }
        for (int i = 0; i < n; i++) {
            int u = tree.seq[i], p = tree.fa[u];
            b[i] = a[i];
            if (p != -1) {
                b[i] = Abel::op(b[tree.in[p]], b[i]);
            }
        }
        sub = Fenwick<Abel>(a);
        path = DualFenwick<Abel>(b);
    }

    X rootQuery(int u) {
        return path.query(tree.in[u]);
```

```
    }

    X pathQuery(int u, int v) {
        int l = hld.lca(u, v);
        X x = rootQuery(l);
        X res = Abel::op(rootQuery(u), rootQuery(v));
        res = Abel::op(res, Abel::inv(Abel::op(x, x)));
        if constexpr (!edge) {
            res = Abel::op(res, sub.query(tree.in[l], tree.in[l] + 1));
        }
        return res;
    }

    X subtreeQuery(int u, int root = -1) {
        if (root == -1) {
            root = tree.root;
        }
        if (root == u) {
            return sub.query(n);
        }
        if (!hld.isAncester(u, root)) {
            return sub.query(tree.in[u] + edge, tree.out[u]);
        }
        int v = hld.jump(root, tree.dep[root] - tree.dep[u] - 1);
        return Abel::op(sub.query(0, tree.in[v]), sub.query(tree.out[v], n));
    }
};
```

3.10.9 树上幺半群

```
template <typename T, typename Mono, bool edge = false>
struct TreeMono {
    using X = typename Mono::Type;

    HLD<T>& hld;
    Tree<T>& tree;
    int n;
    SegmentTree<Mono> seg;
    SegmentTree<RevMono<Mono>> rseg;

    TreeMono(HLD<T>& hld_) : TreeMono(hld_, vector<X>(hld_.tree.n - edge,
Mono::e())) {}
    TreeMono(HLD<T>& hld_, const vector<X>& init) : hld(hld_), tree(hld.tree),
n(tree.n) {
        assert(int(init.size()) == n - edge);
        vector<X> a(n, Mono::e());
        for (int i = 0; i < int(init.size()); i++) {
            a[id(i)] = init[i];
        }
        seg = SegmentTree<Mono>(a);
        rseg = SegmentTree<RevMono<Mono>>(a);
    }

    void modify(int i, const X& x) {
        int p = id(i);
        seg.modify(p, x);
        rseg.modify(p, x);
    }

    void apply(int i, const X& x) {
        int p = id(i);
        modify(i, Mono::op(seg.query(p, p + 1), x));
    }

    X pathQuery(int u, int v) {
        X ans = Mono::e();
        hld.forPath(u, v, edge, [&](int l, int r, bool rev) {
            X cur = (rev ? rseg.query(l, r) : seg.query(l, r));
            ans = Mono::op(ans, cur);
        });
        return ans;
    }

    X subtreeQuery(int u, int root = -1) {
        if (root == -1) {
            root = tree.root;
        }
        if (root == u) {
            return seg.query(0, n);
        }
        if (!hld.isAncester(u, root)) {
            return seg.query(tree.in[u] + edge, tree.out[u]);
        }
        assert(!edge);
        int v = hld.jump(root, tree.dep[root] - tree.dep[u] - 1);
        return Mono::op(seg.query(0, tree.in[v]), seg.query(tree.out[v], n));
    }

    X allQuery() {
        return seg.query(0, n);
    }

```

```
    template <typename F>
    int maxPath(F&& pred, int u, int v) {
        X acc = Mono::e();
        if (edge && !pred(acc)) {
            return -1;
        }

        int ans = v;
        bool done = false;
        hld.forPath(u, v, edge, [&](int l, int r, bool rev) {
            if (done) {
                return;
            }
            X cur = rev ? rseg.query(l, r) : seg.query(l, r);
            X next = Mono::op(acc, cur);
            if (pred(next)) {
                acc = next;
                return;
            }
            int p = (rev ? rseg.minLeft(r, pred, acc) - 1 : seg.maxRight(l, pred,
acc));
            int w = tree.seq[p];
            if constexpr (edge) {
                ans = (rev ? w : tree.fa[w]);
            } else {
                ans = (w == u ? -1 : hld.jump(w, u, 1));
            }
            done = true;
        });
        return ans;
    }

    int id(int i) {
        int u = (edge ? tree.eToV(i) : i);
        return tree.in[u];
    }
};
```

4 网络流

4.1 最大流最小割

```
template <typename T>
struct MaxFlow {
    struct Edge_ {
        int to;
        T cap;
        Edge_(int to_, T cap_) : to(to_), cap(cap_) {}
    };

    int n;
    vector<Edge_> e;
    vector<vector<int>> adj;
    vector<int> cur, level;

    MaxFlow() {}
    MaxFlow(int n_) : n(n_), adj(n), cur(n), level(n) {}

    bool bfs(int s, int t) {
        level.assign(n, -1);
        queue<int> q;
        level[s] = 0;
        q.push(s);
        while (!q.empty()) {
            const int u = q.front();
            q.pop();
            for (auto& i : adj[u]) {
                auto [v, c] = e[i];
                if (c > 0 && level[v] == -1) {
                    level[v] = level[u] + 1;
                    if (v == t) {
                        return true;
                    }
                    q.push(v);
                }
            }
        }
        return false;
    }

    T dfs(int u, int t, T f) {
        if (u == t) {
            return f;
        }
        auto r = f;
        for (int& i = cur[u]; i < int(adj[u].size()); i++) {
            const int& j = adj[u][i];
            auto [v, c] = e[j];
            if (c > 0 && level[v] == level[u] + 1) {
                auto a = dfs(v, t, min(r, c));
```

```
                e[j].cap -= a;
                e[j ^ 1].cap += a;
                r -= a;
                if (r == 0) {
                    return f;
                }
            }
        }
        return f - r;
    }

    void addEdge(int u, int v, T c) {
        adj[u].push_back(e.size());
        e.emplace_back(v, c);
        adj[v].push_back(e.size());
        e.emplace_back(u, 0);
    }

    T flow(int s, int t) {
        T ans = 0;
        while (bfs(s, t)) {
            cur.assign(n, 0);
            ans += dfs(s, t, numeric_limits<T>::max());
        }
        return ans;
    }

    vector<bool> minCut() {
        vector<bool> c(n);
        for (int i = 0; i < n; i++) {
            c[i] = (level[i] != -1);
        }
        return c;
    }

    struct Edge {
        int from;
        int to;
        T cap;
        T flow;
    };

    vector<Edge> edges() {
        vector<Edge> a;
        for (int i = 0; i < e.size(); i += 2) {
            Edge x;
            x.from = e[i + 1].to;
            x.to = e[i].to;
            x.cap = e[i].cap + e[i + 1].cap;
            x.flow = e[i + 1].cap;
            a.push_back(x);
        }
        return a;
    }
};
```

4.2 最小费用最大流

最小费用流的每次增广都要在残量网络上找最短路，而残量网络里有反向边、费用是负的，Dijkstra 直接用不了。势能数组 h 就是用来把边权「平移」成非负的。

做法是把每条边的费用换成约化费用

$$c'(u,v) = c(u,v) + h(u) - h(v),$$

并且每轮求完最短路后令 $h(v) \leftarrow h(v) + \mathrm{dist}'(v)$，其中 dist' 是用约化费用算出的最短路。这样之后的约化费用依然非负—由三角不等式 $\mathrm{dist}'(v) \le \mathrm{dist}'(u) + c'(u,v)$ 整理一下，正好就是新的 $c'(u,v) \ge 0$。

代价是路径长度被平移了常量 $h(s) - h(t)$，但它不影响谁是最短路。代码里每轮累加的是 $h(t)$：因为 $h(s)$ 恒为 0，更新后的 $h(t)$ 恰好等于原网络中 $s \to t$ 的真实最短费用。

初始势能默认全 0，这就要求**初始边权全部非负**；如果原图本来就有负费用边，得根据图的特性自行构建 h 或者直接先跑一遍 SPFA 得到初始 h，再从参数传进来。

```
template <typename T>
struct MinCostFlow {
    struct Edge_ {
        int to;
        T cap;
        T cost;
        Edge_(int to_, T cap_, T cost_) : to(to_), cap(cap_), cost(cost_) {}
    };

    int n;
    vector<Edge_> e;
    vector<vector<int>> adj;
    vector<T> h, dist;
    vector<int> pre;

    bool dijkstra(int s, int t) {
        dist.assign(n, inf<T>);
        pre.assign(n, -1);
        priority_queue<pair<T, int>, vector<pair<T, int>>, greater<pair<T, int>>>
que;
        dist[s] = 0;
        que.emplace(0, s);
```

```
        while (!que.empty()) {
            auto [d, u] = que.top();
            que.pop();
            if (dist[u] != d) {
                continue;
            }
            for (auto& i : adj[u]) {
                int v = e[i].to;
                if (e[i].cap > 0 && dist[v] > d + h[u] - h[v] + e[i].cost) {
                    dist[v] = d + h[u] - h[v] + e[i].cost;
                    pre[v] = i;
                    que.emplace(dist[v], v);
                }
            }
        }
        return dist[t] != inf<T>;
    }

    MinCostFlow() {}
    MinCostFlow(int n_) : n(n_), adj(n), h(n) {}

    void addEdge(int u, int v, T cap, T cost) {
        adj[u].push_back(e.size());
        e.emplace_back(v, cap, cost);
        adj[v].push_back(e.size());
        e.emplace_back(u, 0, -cost);
    }

    void set(const vector<T>& potential) {
        h = potential;
    }

    pair<T, T> flow(int s, int t, T need = inf<T>) {
        T flow = 0;
        T cost = 0;
        while (flow < need && dijkstra(s, t)) {
            for (int i = 0; i < n; i++) {
                if (dist[i] != inf<T>) {
                    h[i] += dist[i];
                }
            }
            T aug = need - flow;
            for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
                aug = min(aug, e[pre[i]].cap);
            }
            for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
                e[pre[i]].cap -= aug;
                e[pre[i] ^ 1].cap += aug;
            }
            flow += aug;
            cost += aug * h[t];
        }
        return {flow, cost};
    }

    struct Edge {
        int from;
        int to;
        T cap;
        T cost;
        T flow;
    };

    vector<Edge> edges() {
        vector<Edge> a;
        for (int i = 0; i < e.size(); i += 2) {
            Edge x;
            x.from = e[i + 1].to;
            x.to = e[i].to;
            x.cap = e[i].cap + e[i + 1].cap;
            x.cost = e[i].cost;
            x.flow = e[i + 1].cap;
            a.push_back(x);
        }
        return a;
    }
};
```

5 数论

5.1 扩展欧几里得算法

```
i64 exgcd(i64 a, i64 b, i64& x, i64& y) {
    if (b == 0) {
        x = 1, y = 0;
        return a;
    }
    i64 d = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return d;
```

```
}

i64 inv(i64 a, i64 m) {
    i64 x, y;
    i64 d = exgcd(a, m, x, y);
    if (d != 1) {
        return -1;
    }
    return (x % m + m) % m;
}

bool solveLinear(i64 a, i64 b, i64 c, i64& x, i64& y) {
    i64 d = exgcd(a, b, x, y);
    if (c % d != 0) {
        return false;
    }
    i64 k = c / d;
    x *= k;
    y *= k;
    return true;
}

pair<i64, i64> solveMod(i64 a, i64 b, i64 m) {
    assert(m > 0);
    b *= -1;
    i64 x, y;
    i64 g = exgcd(a, m, x, y);
    if (g < 0) {
        g *= -1;
        x *= -1;
        y *= -1;
    }
    if (b % g != 0) {
        return {-1, -1};
    }
    x = x * (b / g) % (m / g);
    if (x < 0) {
        x += m / g;
    }
    return {x, m / g};
}
```

5.2 线性筛

```
vector<int> minp, primes;

void sieve(int n) {
    minp.assign(n + 1, 0);
    primes.clear();

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            primes.push_back(i);
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                break;
            }
        }
    }
}
```

5.3 莫比乌斯反演

莫比乌斯反演用来在「按约数求和」和「按倍数求和」之间互转。

$\mu(n)$ 的定义：$n=1$ 时为 1；n 是 k 个互异素数之积时为 $(-1)^k$；含平方因子时为 0。一切从这条恒等式出发：

$$\sum_{d\mid n}\mu(d)=[n=1].$$

于是有两种方向的反演：

$$f(n)=\sum_{d\mid n}g(d)\Longrightarrow g(n)=\sum_{d\mid n}\mu(d)f\left(\frac{n}{d}\right),$$

$$f(n)=\sum_{n\mid d}g(d)\Longrightarrow g(n)=\sum_{n\mid d}\mu\left(\frac{d}{n}\right)f(d).$$

实际做题更常用的是「换条件」的用法：把 $[\gcd(i,j)=1]$ 换成 $\sum_{d\mid\gcd(i,j)}\mu(d)$，再交换求和顺序，条件就从 gcd 挪到了 d：

$$\sum_{i=1}^{n}\sum_{j=1}^{m}[\gcd(i,j)=1]=\sum_{d=1}^{\min(n,m)}\mu(d)\left\lfloor\frac{n}{d}\right\rfloor\left\lfloor\frac{m}{d}\right\rfloor.$$

右边只要 μ 的前缀和就能整除分块，μ 用下面的 `sieve()` 线性筛出来。

```
vector<int> minp, primes, mu;

void sieve(int n) {
    minp.assign(n + 1, 0);
```

```
    mu.assign(n + 1, 0);
    primes.clear();
    mu[1] = 1;

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            mu[i] = -1;
            primes.push_back(i);
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                mu[i * p] = 0;
                break;
            } else {
                mu[i * p] = -mu[i];
            }
        }
    }
}

```

5.4 欧拉函数

$$\varphi(n) = n\prod_{i=1}^{k}\left(1-\frac{1}{p_i}\right).$$

两条常用性质：它是积性的，即 $\gcd(a,b)=1$ 时 $\varphi(ab)=\varphi(a)\varphi(b)$；并且 $\sum_{d|n}\varphi(d)=n$，也即 $\varphi(n)=\sum_{d|n}\mu(d)\frac{n}{d}$。

欧拉定理：$\gcd(a,m)=1$ 时 $a^{\varphi(m)}\equiv 1 (\bmod m)$。竞赛里更常用的是扩展欧拉定理，它给指数降幂—$b\geq\varphi(m)$ 时对任意 a 都有

$$a^b \equiv a^{b \bmod \varphi(m)+\varphi(m)} (\bmod m).$$

线性筛求 φ 的递推：素数的 $\varphi(p)=p-1$；对 $i\cdot p$，若 $p\mid i$ 则 $\varphi(ip)=p\varphi(i)$，否则 $\varphi(ip)=(p-1)\varphi(i)$。见下面的 `sieve()`；只求单个数用 `Phi()`，复杂度 $O(\sqrt{n})$。

```
vector<int> minp, primes, phi;

// 线性筛（用于求解全部数的欧拉函数）
void sieve(int n) {
    minp.assign(n + 1, 0);
    phi.assign(n + 1, 0);
    primes.clear();
    phi[1] = 1;

    for (int i = 2; i <= n; i++) {
        if (minp[i] == 0) {
            minp[i] = i;
            primes.push_back(i);
            phi[i] = i - 1;
        }

        for (auto p : primes) {
            if (i * p > n) {
                break;
            }
            minp[i * p] = p;
            if (p == minp[i]) {
                phi[i * p] = phi[i] * p;
                break;
            } else {
                phi[i * p] = phi[i] * (p - 1);
            }
        }
    }
}

// 求解单个数的欧拉函数
int Phi(int n) {
    int res = n;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            while (n % i == 0) {
                n /= i;
            }
            res = res / i * (i - 1);
        }
    }
    if (n > 1) {
        res = res / n * (n - 1);
    }
    return res;
}

```

5.5 Miller-Rabin

```
bool isPrime(i64 n) {
    if (n < 2) {
```

```
        return false;
    }
    constexpr array<int, 10> smallPrimes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29};
    for (auto& p : smallPrimes) {
        if (n % p == 0) {
            return n == p;
        }
    }
    if (n < 31 * 31) {
        return true;
    }
    constexpr array<int, 7> bases = {2, 325, 9375, 28178, 450775, 9780504,
1795265022};
    int s = __builtin_ctzll(n - 1);
    i64 d = (n - 1) >> s;
    for (auto& a : bases) {
        if (a % n == 0) {
            continue;
        }
        i64 cur = power(a, d, n);
        if (cur == 1) {
            continue;
        }
        bool witness = true;
        for (int r = 0; r < s; r++) {
            if (cur == n - 1) {
                witness = false;
                break;
            }
            cur = mul(cur, cur, n);
        }
        if (witness) {
            return false;
        }
    }
    return true;
}
```

5.6 Pollard-Rho

```
i64 rho(i64 n, i64 c) {
    auto f = [&](i64 x) {
        return static_cast<i64>((static_cast<i128>(x) * x + c) % n);
    };
    i64 x = 1, y = 2, z = 1, q = 1;
    i64 g = 1;
    constexpr i64 m = 127;
    for (i64 r = 1; g == 1; r <<= 1) {
        x = y;
        for (i64 i = 0; i < r; i++) {
            y = f(y);
        }
        for (i64 k = 0; k < r && g == 1; k += m) {
            z = y;
            for (i64 i = 0; i < min(m, r - k); i++) {
                y = f(y);
                q = mul(q, abs(x - y), n);
            }
            g = gcd(q, n);
        }
    }
    if (g == n) {
        do {
            z = f(z);
            g = gcd(abs(x - z), n);
        } while (g == 1);
    }
    return g;
}

i64 primeFactor(i64 n) {
    assert(n >= 2);
    if (n % 2 == 0) {
        return 2;
    }
    while (!isPrime(n)) {
        n = rho(n, randInt(1, n));
    }
    return n;
}

vector<pair<i64, int>> factorize(i64 n) {
    vector<pair<i64, int>> pf;
    for (int p = 2; p < 100; p++) {
        if (p * p > n) {
            break;
        }
        if (n % p == 0) {
            int e = 0;
```

```
            do {
                n /= p;
                e++;
            } while (n % p == 0);
            pf.emplace_back(p, e);
        }
    }
    while (n > 1) {
        i64 p = primeFactor(n);
        int e = 0;
        do {
            n /= p;
            e++;
        } while (n % p == 0);
        pf.emplace_back(p, e);
    }
    sort(pf.begin(), pf.end());
    return pf;
}
```

5.7 约数构造

```
vector<i64> buildDivisors(const vector<pair<i64, int>>& factors) {
    vector<i64> divisors = {1};
    for (auto& [p, e] : factors) {
        int sz = divisors.size();
        for (int i = 0; i < sz; i++) {
            i64 x = divisors[i];
            for (int j = 0; j < e; j++) {
                x *= p;
                divisors.push_back(x);
            }
        }
    }
    return divisors;
}
```

6 组合数学

6.1 组合数

```
template <typename T>
struct Combination {
    int n;
    vector<T> fact_, invfact_, inv_;

    Combination() : n(0), fact_{1}, invfact_{1}, inv_{0} {}
    Combination(int n_) : Combination() {
        init(n_);
    }

    void init(int n_) {
        if (n_ <= n) {
            return;
        }
        fact_.resize(n_ + 1);
        invfact_.resize(n_ + 1);
        inv_.resize(n_ + 1);

        for (int i = n + 1; i <= n_; i++) {
            fact_[i] = fact_[i - 1] * i;
        }
        invfact_[n_] = fact_[n_].inv();
        for (int i = n_; i > n; i--) {
            invfact_[i - 1] = invfact_[i] * i;
            inv_[i] = invfact_[i] * fact_[i - 1];
        }
        n = n_;
    }

    T fact(int k) {
        if (k > n) {
            init(2 * k);
        }
        return fact_[k];
    }

    T invfact(int k) {
        if (k > n) {
            init(2 * k);
        }
        return invfact_[k];
    }

    T inv(int k) {
        if (k > n) {
            init(2 * k);
        }
        return inv_[k];
    }

    T binom(int n, int m) {
```

```
        if (n < m || m < 0) {
            return 0;
        }
        return fact(n) * invfact(m) * invfact(n - m);
    }

    T perm(int n, int m) {
        if (n < m || m < 0) {
            return 0;
        }
        return fact(n) * invfact(n - m);
    }

    T multicomb(int n, int m) {
        if (n < 0 || m < 0) {
            return 0;
        }
        if (n == 0) {
            return (m == 0);
        }
        return binom(n + m - 1, m);
    }

    T Catalan(int n) {
        if (n < 0) {
            return 0;
        }
        return binom(2 * n, n) * inv(n + 1);
    }
};
```

6.2 康托展开

```
vector<int> toLehmer(vector<int>& p) {
    int n = p.size();
    Fenwick<AddMono<int>> fen(n);
    vector<int> c(n);
    for (int i = n - 1; i >= 0; i--) {
        c[i] = fen.query(p[i]);
        fen.add(p[i], 1);
    }
    return c;
}

vector<int> fromLehmer(vector<int>& c) {
    int n = c.size();
    vector<int> p(n);
    Fenwick<AddMono<int>> fen(n);
    for (int i = 0; i < n; i++) {
        fen.add(i, 1);
    }
    for (int i = 0; i < n; i++) {
        p[i] = fen.kth(c[i]);
        fen.add(p[i], -1);
    }
    return p;
}

i64 toRank(vector<int>& p) {
    int n = p.size();
    auto c = toLehmer(p);
    i64 fac = 1, rk = 0;
    for (int i = n - 1; i >= 0; i--) {
        rk += c[i] * fac;
        fac *= (n - i);
    }
    return rk;
}

vector<int> fromRank(int n, i64 rk) {
    vector<int> c(n);
    for (int i = 1; i <= n; i++) {
        c[n - i] = rk % i;
        rk /= i;
    }
    return fromLehmer(c);
}
```

6.3 逆序对

```
i64 invCount(vector<i64>& a, int l, int r, vector<i64>& tmp) {
    if (r - l <= 1) return 0;

    i64 res = 0;
    int mid = (l + r) / 2;
    res += invCount(a, l, mid, tmp);
    res += invCount(a, mid, r, tmp);

    int i = l, j = mid, k = l;
    while (i < mid && j < r) {
        if (a[i] <= a[j]) {
            tmp[k++] = a[i++];
```

```
        } else {
            tmp[k++] = a[j++];
            res += mid - i;
        }
    }
    while (i < mid) {
        tmp[k++] = a[i++];
    }
    while (j < r) {
        tmp[k++] = a[j++];
    }

    for (i = l; i < r; i++) {
        a[i] = tmp[i];
    }

    return res;
}

```

6.4 容斥原理

```
// cap(S) 表示 S 集合的交集大小

// 交集形式
template <typename T, typename F>
T ie(int n, F&& cap) {
    T ans{};
    for (int S = 1; S < (1 << n); S++) {
        T cur = cap(S);
        if (popcount(u32(S)) & 1) {
            ans += cur;
        } else {
            ans -= cur;
        }
    }
    return ans;
}

// 补集形式
template <typename T, typename F>
T ieComp(int n, F&& cap) {
    T ans{};
    for (int S = 0; S < (1 << n); S++) {
        T cur = cap(S);
        if (popcount(u32(S)) & 1) {
            ans -= cur;
        } else {
            ans += cur;
        }
    }
    return ans;
}

// 恰好满足 k 个性质的元素数量
template <typename T, typename F>
vector<T> ieExact(int n, F&& cap) {
    vector<T> N(n + 1, T{});
    for (int S = 0; S < (1 << n); S++) {
        int k = popcount(u32(S));
        N[k] += cap(S);
    }

    vector<T> binom(n + 2, T{}), E(n + 1, T{});
    binom[0] = T{1};
    for (int j = 0; j <= n; j++) {
        for (int k = 0; k <= j; k++) {
            T term = N[j] * binom[k];
            if ((j - k) & 1) {
                E[k] -= term;
            } else {
                E[k] += term;
            }
        }
        for (int k = j + 1; k >= 1; k--) {
            binom[k] += binom[k - 1];
        }
    }
    return E;
}

// 至少满足 k 个
template <typename T, typename F>
vector<T> ieLeast(int n, F&& cap) {
    auto E = ieExact<T>(n, cap);
    for (int k = n - 1; k >= 0; k--) {
        E[k] += E[k + 1];
    }
    return E;
}
```

```

// 至多满足 k 个
template <typename T, typename F>
vector<T> ieMost(int n, F&& cap) {
    auto E = ieExact<T>(n, cap);
    for (int k = 0; k < n; k++) {
        E[k + 1] += E[k];
    }
    return E;
}
```

6.5 zeta 变换

```
// 子集和：f[mask] = sum_{sub ⊆ mask} f[sub]
// 方向：小的 mask → 大的 mask（"吸收"不带的位）
template <typename T>
void subsetZeta(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask] += f[mask ^ i];
            }
        }
    }
}

// 子集和的逆
template <typename T>
void subsetMobius(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask] -= f[mask ^ i];
            }
        }
    }
}

// 超集和：f[mask] = sum_{sup ⊇ mask} f[sup]
// 方向：大的 mask → 小的 mask（"分发"给不带这一位的 mask）
template <typename T>
void supersetZeta(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask ^ i] += f[mask];
            }
        }
    }
}

// 超集和的逆
template <typename T>
void supersetMobius(vector<T>& f) {
    int n = f.size();
    for (int i = 1; i < n; i <<= 1) {
        for (int mask = 0; mask < n; mask++) {
            if (mask & i) {
                f[mask ^ i] -= f[mask];
            }
        }
    }
}
```

6.6 Polyomino

```
struct Polyomino {
    using Cell = pair<int, int>;
    set<Cell> cells;
    Polyomino() {}
    Polyomino(const set<Cell>& cells_) : cells(cells_) {}
};

Polyomino normalize(const Polyomino& p) {
    if (p.cells.empty()) {
        return p;
    }
    int minX = p.cells.begin()->first;
    int minY = p.cells.begin()->second;
    for (auto& [x, y] : p.cells) {
        minX = min(minX, x);
        minY = min(minY, y);
    }
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(x - minX, y - minY);
    }
```

```
    return res;
}

Polyomino rotate(const Polyomino& p) {
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(y, -x);
    }
    return normalize(res);
}

Polyomino flip(const Polyomino& p) {
    Polyomino res;
    for (auto& [x, y] : p.cells) {
        res.cells.emplace(x, -y);
    }
    return normalize(res);
}
```

7 数据结构

7.1 线段树

7.1.1 线段树

```
template <typename Mono>
struct SegmentTree {
    using X = typename Mono::Type;

    int n;
    vector<X> info;

    SegmentTree() : n(0) {}
    SegmentTree(int n_) : SegmentTree(vector<X>(n_, Mono::e())) {}
    SegmentTree(const vector<X>& init) : n(init.size()) {
        int logn = __lg(n);
        info.assign(4 << logn, Mono::e());
        auto build = [&](auto&& self, int p, int l, int r) -> void {
            if (r - l == 1) {
                info[p] = init[l];
                return;
            }
            int mid = (l + r) / 2;
            self(self, 2 * p, l, mid);
            self(self, 2 * p + 1, mid, r);
            pull(p);
        };
        build(build, 1, 0, n);
    }

    void pull(int p) {
        info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
    }

    void modify(int p, int l, int r, int x, const X& v) {
        if (r - l == 1) {
            info[p] = v;
            return;
        }
        int mid = (l + r) / 2;
        if (x < mid) {
            modify(2 * p, l, mid, x, v);
        } else {
            modify(2 * p + 1, mid, r, x, v);
        }
        pull(p);
    }
    void modify(int p, const X& v) {
        modify(1, 0, n, p, v);
    }

    X query(int p, int l, int r, int x, int y) {
        if (y <= l || r <= x) {
            return Mono::e();
        }
        if (x <= l && r <= y) {
            return info[p];
        }
        int mid = (l + r) / 2;
        return Mono::op(query(2 * p, l, mid, x, y), query(2 * p + 1, mid, r, x, y));
    }
    X query(int l, int r) {
        return query(1, 0, n, l, r);
    }

    template <typename F>
    int findFirst(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
```

```
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        int res = findFirst(2 * p, l, mid, x, y, pred);
        if (res == -1) {
            res = findFirst(2 * p + 1, mid, r, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findFirst(int l, int r, F&& pred) {
        return findFirst(1, 0, n, l, r, pred);
    }

    template <typename F>
    int findLast(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        int res = findLast(2 * p + 1, mid, r, x, y, pred);
        if (res == -1) {
            res = findLast(2 * p, l, mid, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findLast(int l, int r, F&& pred) {
        return findLast(1, 0, n, l, r, pred);
    }

    template <typename F>
    int maxRight(int l, F&& pred, X& sum) {
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
    int minLeft(int r, F&& pred, X& sum) {
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.1.2 懒标记线段树

```
template <typename ActMono>
struct LazySegmentTree {
    using Mono = typename ActMono::Mono;
    using Tag = typename ActMono::Tag;
    using X = typename ActMono::X;
    using Y = typename ActMono::Y;

    int n;
    vector<X> info;
    vector<Y> tag;

    LazySegmentTree() : n(0) {}
    LazySegmentTree(int n_) : LazySegmentTree(vector<X>(n_, Mono::e())) {}
    LazySegmentTree(const vector<X>& init) : n(init.size()) {
        int logn = __lg(n);
        info.assign(4 << logn, Mono::e());
        tag.assign(4 << logn, Tag::e());
        auto build = [&](auto&& self, int p, int l, int r) -> void {
            if (r - l == 1) {
                info[p] = init[l];
                return;
            }
```

```
            int mid = (l + r) / 2;
            self(self, 2 * p, l, mid);
            self(self, 2 * p + 1, mid, r);
            pull(p);
        };
        build(build, 1, 0, n);
    }

    void pull(int p) {
        info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
    }

    void update(int p, int len, const Y& v) {
        info[p] = ActMono::act(info[p], v, len);
        tag[p] = Tag::op(tag[p], v);
    }

    void push(int p, int len) {
        update(2 * p, len / 2, tag[p]);
        update(2 * p + 1, (len + 1) / 2, tag[p]);
        tag[p] = Tag::e();
    }

    void modify(int p, int l, int r, int x, const X& v) {
        if (r - l == 1) {
            info[p] = v;
            return;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        if (x < mid) {
            modify(2 * p, l, mid, x, v);
        } else {
            modify(2 * p + 1, mid, r, x, v);
        }
        pull(p);
    }
    void modify(int p, const X& v) {
        modify(1, 0, n, p, v);
    }

    X query(int p, int l, int r, int x, int y) {
        if (y <= l || r <= x) {
            return Mono::e();
        }
        if (x <= l && r <= y) {
            return info[p];
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        return Mono::op(query(2 * p, l, mid, x, y), query(2 * p + 1, mid, r, x, y));
    }
    X query(int l, int r) {
        return query(1, 0, n, l, r);
    }

    void apply(int p, int l, int r, int x, int y, const Y& v) {
        if (y <= l || r <= x) {
            return;
        }
        if (x <= l && r <= y) {
            update(p, r - l, v);
            return;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        apply(2 * p, l, mid, x, y, v);
        apply(2 * p + 1, mid, r, x, y, v);
        pull(p);
    }
    void apply(int l, int r, const Y& v) {
        return apply(1, 0, n, l, r, v);
    }

    template <typename F>
    int findFirst(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findFirst(2 * p, l, mid, x, y, pred);
```

```
        if (res == -1) {
            res = findFirst(2 * p + 1, mid, r, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findFirst(int l, int r, F&& pred) {
        return findFirst(1, 0, n, l, r, pred);
    }

    template <typename F>
    int findLast(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findLast(2 * p + 1, mid, r, x, y, pred);
        if (res == -1) {
            res = findLast(2 * p, l, mid, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findLast(int l, int r, F&& pred) {
        return findLast(1, 0, n, l, r, pred);
    }

    template <typename F>
    int maxRight(int l, F&& pred) {
        X sum = Mono::e();
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
    int minLeft(int r, F&& pred) {
        X sum = Mono::e();
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.1.3 动态开点线段树

```
template <typename ActMono>
struct DynamicSegmentTree {
    using Mono = typename ActMono::Mono;
    using Tag = typename ActMono::Tag;
    using X = typename ActMono::X;
    using Y = typename ActMono::Y;

    int n;
    int root;
    vector<X> info;
    vector<Y> tag;
    vector<int> lc, rc;

    DynamicSegmentTree() : n(0), root(-1) {}
    DynamicSegmentTree(int n_) : n(n_), root(-1) {
        root = newNode();
    }
    DynamicSegmentTree(const vector<X>& init) : n(init.size()), root(-1) {
        auto build = [&](auto&& self, int l, int r) -> int {
            int p = newNode();
            if (r - l == 1) {
                info[p] = init[l];
                return p;
            }
            int mid = (l + r) / 2;
```

```
            lc[p] = self(self, l, mid);
            rc[p] = self(self, mid, r);
            pull(p);
            return p;
        };
        root = build(build, 0, n);
    }

    int newNode() {
        info.push_back(Mono::e());
        tag.push_back(Tag::e());
        lc.push_back(-1);
        rc.push_back(-1);
        return int(info.size()) - 1;
    }

    void pull(int p) {
        info[p] = Mono::op(info[lc[p]], info[rc[p]]);
    }

    void update(int p, int len, const Y& v) {
        info[p] = ActMono::act(info[p], v, len);
        tag[p] = Tag::op(tag[p], v);
    }

    void push(int p, int len) {
        if (lc[p] == -1) {
            lc[p] = newNode();
        }
        if (rc[p] == -1) {
            rc[p] = newNode();
        }
        update(lc[p], len / 2, tag[p]);
        update(rc[p], (len + 1) / 2, tag[p]);
        tag[p] = Tag::e();
    }

    void modify(int p, int l, int r, int x, const X& v) {
        if (r - l == 1) {
            info[p] = v;
            return;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        if (x < mid) {
            modify(lc[p], l, mid, x, v);
        } else {
            modify(rc[p], mid, r, x, v);
        }
        pull(p);
    }
    void modify(int x, const X& v) {
        modify(root, 0, n, x, v);
    }

    X query(int p, int l, int r, int x, int y) {
        if (y <= l || r <= x) {
            return Mono::e();
        }
        if (x <= l && r <= y) {
            return info[p];
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        return Mono::op(query(lc[p], l, mid, x, y), query(rc[p], mid, r, x, y));
    }
    X query(int l, int r) {
        return query(root, 0, n, l, r);
    }

    void apply(int p, int l, int r, int x, int y, const Y& v) {
        if (y <= l || r <= x) {
            return;
        }
        if (x <= l && r <= y) {
            update(p, r - l, v);
            return;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        apply(lc[p], l, mid, x, y, v);
        apply(rc[p], mid, r, x, y, v);
        pull(p);
    }
    void apply(int l, int r, const Y& v) {
        apply(root, 0, n, l, r, v);
    }

    template <typename F>
```

```
    int findFirst(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findFirst(lc[p], l, mid, x, y, pred);
        if (res == -1) {
            res = findFirst(rc[p], mid, r, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findFirst(int l, int r, F&& pred) {
        return findFirst(root, 0, n, l, r, pred);
    }

    template <typename F>
    int findLast(int p, int l, int r, int x, int y, F&& pred) {
        if (y <= l || r <= x) {
            return -1;
        }
        if (x <= l && r <= y && !pred(info[p])) {
            return -1;
        }
        if (r - l == 1) {
            return l;
        }
        int mid = (l + r) / 2;
        push(p, r - l);
        int res = findLast(rc[p], mid, r, x, y, pred);
        if (res == -1) {
            res = findLast(lc[p], l, mid, x, y, pred);
        }
        return res;
    }
    template <typename F>
    int findLast(int l, int r, F&& pred) {
        return findLast(root, 0, n, l, r, pred);
    }

    template <typename F>
    int maxRight(int l, F&& pred, X& sum) {
        int p = findFirst(l, n, [&](const X& x) {
            X v = Mono::op(sum, x);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p == -1 ? n : p;
    }

    template <typename F>
    int minLeft(int r, F&& pred, X& sum) {
        int p = findLast(0, r, [&](const X& x) {
            X v = Mono::op(x, sum);
            if (!pred(v)) {
                return true;
            }
            sum = v;
            return false;
        });
        return p + 1;
    }
};
```

7.2 树状数组

7.2.1 树状数组

```
template <typename Mono>
struct Fenwick {
    using X = typename Mono::Type;

    int n;
    vector<X> a;

    Fenwick(int n_ = 0) : n(n_), a(n, Mono::e()) {}
    Fenwick(const vector<X>& init) : n(init.size()), a(init) {
        for (int i = 1; i <= n; i++) {
            int j = i + (i & -i);
            if (j <= n) {
                a[j - 1] = Mono::op(a[j - 1], a[i - 1]);
```

```
            }
        }
    }

    void add(int x, const X& v) {
        assert(0 <= x && x < n);
        for (int i = x + 1; i <= n; i += i & -i) {
            a[i - 1] = Mono::op(a[i - 1], v);
        }
    }

    X query(int x) {
        assert(0 <= x && x <= n);
        X res = Mono::e();
        for (int i = x; i > 0; i -= i & -i) {
            res = Mono::op(a[i - 1], res);
        }
        return res;
    }

    X query(int l, int r) {
        assert(l <= r);
        return Mono::op(Mono::inv(query(l)), query(r));
    }

    // lower_bound
    int kth(X k) {
        int x = 0;
        for (int i = 1 << __lg(n); i; i /= 2) {
            if (x + i <= n && k >= a[x + i - 1]) {
                x += i;
                k -= a[x - 1];
            }
        }
        return x;
    }
};
```

7.2.2 区间加、单点查树状数组

```
template <typename Abel>
struct DualFenwick {
    using X = typename Abel::Type;

    int n;
    Fenwick<Abel> fen;

    DualFenwick(int n_ = 0) : n(n_), fen(n) {}
    DualFenwick(const vector<X>& init) : n(init.size()) {
        vector<X> d(n);
        X pre = Abel::e();
        for (int i = 0; i < n; i++) {
            d[i] = Abel::op(Abel::inv(pre), init[i]);
            pre = init[i];
        }
        fen = Fenwick<Abel>(d);
    }

    void add(int l, int r, const X& v) {
        assert(0 <= l && l < r && r <= n);
        fen.add(l, v);
        if (r < n) {
            fen.add(r, Abel::inv(v));
        }
    }

    X query(int x) {
        assert(0 <= x && x < n);
        return fen.query(x + 1);
    }
};
```

7.2.3 区间加、区间和树状数组

```
template <typename Abel>
struct FenwickRangeAdd {
    using X = typename Abel::Type;

    int n;
    Fenwick<Abel> fen0, fen1;

    FenwickRangeAdd(int n_ = 0) : n(n_), fen0(n + 1), fen1(n + 1) {}
    FenwickRangeAdd(const vector<X>& init) : n(init.size()) {
        vector<X> d(n + 1), w(n + 1);
        X pre = Abel::e();
        for (int i = 0; i <= n; i++) {
            X cur = (i < n ? init[i] : Abel::e());
            d[i] = Abel::op(Abel::inv(pre), cur);
            w[i] = Abel::pow(d[i], -i);
            pre = cur;
        }
        fen0 = Fenwick<Abel>(w);
```

```
        fen1 = Fenwick<Abel>(d);
    }

    void add(int l, int r, const X& v) {
        assert(0 <= l && l < r && r <= n);
        fen0.add(l, Abel::pow(v, -l));
        fen0.add(r, Abel::pow(v, r));
        fen1.add(l, v);
        fen1.add(r, Abel::inv(v));
    }

    X query(int r) {
        assert(0 <= r && r <= n);
        return Abel::op(Abel::pow(fen1.query(r), r), fen0.query(r));
    }

    X query(int l, int r) {
        assert(0 <= l && l <= r && r <= n);
        return Abel::op(Abel::inv(query(l)), query(r));
    }
};
```

7.2.4 二维树状数组

```
template <typename Mono>
struct Fenwick2D {
    using X = typename Mono::Type;

    int h, w;
    vector<X> a;

    Fenwick2D(int h_ = 0, int w_ = 0) : h(h_), w(w_), a((h + 1) * (w + 1), Mono::e()) {}

    void add(int x, int y, const X& v) {
        assert(0 <= x && x < h);
        assert(0 <= y && y < w);
        for (int i = x + 1; i <= h; i += i & -i) {
            for (int j = y + 1; j <= w; j += j & -j) {
                a[i * (w + 1) + j] = Mono::op(a[i * (w + 1) + j], v);
            }
        }
    }

    X query(int x, int y) {
        assert(0 <= x && x <= h);
        assert(0 <= y && y <= w);
        X res = Mono::e();
        for (int i = x; i > 0; i -= i & -i) {
            for (int j = y; j > 0; j -= j & -j) {
                res = Mono::op(res, a[i * (w + 1) + j]);
            }
        }
        return res;
    }

    X query(int x1, int x2, int y1, int y2) {
        assert(0 <= x1 && x1 <= x2 && x2 <= h);
        assert(0 <= y1 && y1 <= y2 && y2 <= w);
        X s22 = query(x2, y2);
        X s12 = query(x1, y2);
        X s21 = query(x2, y1);
        X s11 = query(x1, y1);
        return Mono::op(Mono::op(s22, Mono::inv(s12)), Mono::op(Mono::inv(s21), s11));
    }
};
```

7.3 PBDS

7.3.1 哈希表

HashMap<K, V, Hash> / HashSet<K, Hash>。

insert({k, v}) / insert(k) 插入，返回（迭代器，是否成功）；erase(k) 删除并返回是否存在；find(k) 查找，未找到返回 end()；Map 的 a[k] 访问值，键不存在时插入默认值。

上述操作期望 $O(1)$（插入、删除为均摊），最坏 $O(n)$；clear() 为 $O(n)$。

```
struct splitmix64Hash {
    static u64 splitmix64(u64 x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(u64 x) const {
        static const u64 FIXED_RANDOM = chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

template <typename K, typename V, typename Hash = splitmix64Hash>
using HashMap = __gnu_pbds::gp_hash_table<K, V, Hash>;
```

```

template <typename K, typename Hash = splitmix64Hash>
using HashSet = HashMap<K, __gnu_pbds::null_type, Hash>;
```

7.3.2 红黑树

`OrderedSet<K, Comp>` / `OrderedMap<K, V, Comp>`：键唯一，默认升序；下述大小关系按 `Comp`。

`insert` 插入，`erase(k)` 删除，`find(k)` 查找；`lower_bound(k)` / `upper_bound(k)` 返回首个 $\geq k$ / $> k$ 的迭代器；`Map` 的 `a[k]` 访问或插入默认值。均为 $O(\log n)$。

`find_by_order(k)` 返回从 0 计的第 k 个元素的迭代器，越界返回 `end()`；`order_of_key(k)` 返回严格排在 k 前的键数。均为 $O(\log n)$。

`a.split(k, b)`：a 保留 $\leq k$，$> k$ 移入 b，清空旧 b；$O(n+m)$，其中 m 为旧 b 大小。

`a.join(b)`：合并并清空 b，要求一棵树的所有键均小于另一棵树的所有键；$O(\log(n+m))$。

```
template <typename K, typename V, typename Comp = less<K>>
using OrderedMap = __gnu_pbds::tree<
    K, V, Comp,
    __gnu_pbds::rb_tree_tag,
    __gnu_pbds::tree_order_statistics_node_update
>;

template <typename K, typename Comp = less<K>>
using OrderedSet = OrderedMap<K, __gnu_pbds::null_type, Comp>;
```

7.3.3 配对堆

`PairingHeap<T, Comp>`：默认大根堆，`greater<T>` 为小根堆。

`push(x)` 插入并返回句柄；`top()` 取堆顶；`pop()` 删除堆顶；`modify(it, x)` 修改句柄对应元素；`erase(it)` 删除元素。修改不使句柄失效，删除使其失效。

`a.join(b)` 合并并清空 b；`a.split(pred, b)` 将满足 pred 的元素移入 b，清空旧 b；`erase_if(pred)` 删除满足 pred 的元素并返回数量。

`top`、`push`、`join`：$O(1)$；`pop`、`modify`、`erase`：均摊 $O(\log n)$；`erase_if`、`clear`：$O(n)$；`split`：$O(n+m)$，m 为旧 b 大小。

```
template <typename T, typename Comp = less<T>>
using PairingHeap = __gnu_pbds::priority_queue<T, Comp,
__gnu_pbds::pairing_heap_tag>;
```

7.4 倍增

```
template <typename Mono>
struct BinaryLifting {
    using X = typename Mono::Type;

    int n, logn;
    vector<vector<int>> nxt;
    vector<vector<X>> info;

    BinaryLifting() : n(0) {}
    BinaryLifting(int n_) : n(n_), logn(__lg(n)) {
        nxt.assign(logn + 1, vector<int>(n, -1));
        info.assign(logn + 1, vector<X>(n, Mono::e()));
    }

    void add(int u, int v, const X& x) {
        assert(-1 <= v && v < n);
        nxt[0][u] = v;
        info[0][u] = x;
    }

    void build() {
        for (int k = 0; k < logn; k++) {
            for (int u = 0; u < n; u++) {
                int v = nxt[k][u];
                if (v == -1) {
                    nxt[k + 1][u] = -1;
                    info[k + 1][u] = info[k][u];
                } else {
                    nxt[k + 1][u] = nxt[k][v];
                    info[k + 1][u] = Mono::op(info[k][u], info[k][v]);
                }
            }
        }
    }

    pair<int, X> query(int u, i64 step) {
        assert(0 <= step && step < (1LL << (logn + 1)));
        X res = Mono::e();
        for (int k = logn; k >= 0 && u != -1; k--) {
            if ((step >> k) & 1) {
                res = Mono::op(res, info[k][u]);
                u = nxt[k][u];
            }
        }
        return {u, res};
    }

    int jump(int u, i64 step) {
        assert(0 <= step && step < (1LL << (logn + 1)));
        for (int k = logn; k >= 0 && u != -1; k--) {
            if ((step >> k) & 1) {
                u = nxt[k][u];
            }
        }
        return u;
```

```
    }
};
```

7.5 并查集

```
struct DSU {
    vector<int> f, siz;

    DSU() {}
    DSU(int n) {
        f.resize(n);
        iota(f.begin(), f.end(), 0);
        siz.assign(n, 1);
    }

    int find(int x) {
        while (x != f[x]) {
            x = f[x] = f[f[x]];
        }
        return x;
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool merge(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            return false;
        }
        siz[x] += siz[y];
        f[y] = x;
        return true;
    }

    int size(int x) {
        return siz[find(x)];
    }
};
```

7.6 稀疏表

```
template <typename Mono>
struct SparseTable {
    using X = typename Mono::Type;

    int n;
    vector<vector<X>> st;

    SparseTable() : n(0) {}
    SparseTable(const vector<X>& a) : n(a.size()) {
        int logn = __lg(n);
        st.assign(logn + 1, vector<X>(n));
        for (int i = 0; i < n; i++) {
            st[0][i] = a[i];
        }
        for (int j = 1; (1 << j) <= n; j++) {
            for (int i = 0; i + (1 << j) <= n; i++) {
                st[j][i] = Mono::op(st[j - 1][i], st[j - 1][i + (1 << (j - 1))]);
            }
        }
    }

    X query(int l, int r) {
        assert(l < r);
        int len = r - l;
        int t = __lg(len);
        return Mono::op(st[t][l], st[t][r - (1 << t)]);
    }
};
```

7.7 01-Trie

```
struct BinaryTrie {
    struct Node {
        array<int, 2> ch;
        int end;
        Node() {
            ch[0] = ch[1] = 0;
            end = 0;
        }
    };

    vector<Node> tr;
    int tot;

    BinaryTrie(int n = 0) : tr(n), tot(1) {}

    void insert(int x) {
        int p = 1;
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
```

```
            if (!tr[p].ch[c]) {
                ++tot;
                tr[p].ch[c] = tot;
            }
            p = tr[p].ch[c];
            tr[p].end++;
        }
    }

    int query(int x) {
        int p = 1, res = 0;
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
            if (tr[p].ch[c ^ 1]) {
                res |= (1 << i);
                p = tr[p].ch[c ^ 1];
            } else {
                p = tr[p].ch[c];
            }
        }
        return res;
    }

    void erase(int x) {
        int p = 1;
        for (int i = 30; i >= 0; i--) {
            int c = x >> i & 1;
            int nxt = tr[p].ch[c];
            tr[nxt].end--;
            if (tr[nxt].end == 0) {
                 tr[p].ch[c] = 0;
            }
            p = nxt;
        }
    }
};
```

7.8 笛卡尔树

```
template <typename T, bool IS_MIN = false>
struct CartesianTree {
    int n;
    vector<T>& a;
    vector<int> L, R, p;
    // range[i]: the interval of the subtree rooted at i
    vector<pair<int, int>> range;
    int root;

    CartesianTree(vector<T>& a_) : n(a_.size()), a(a_), root(-1) {
        L.assign(n, -1);
        R.assign(n, -1);
        p.assign(n, -1);
        range.assign(n, {0, n});
        vector<int> stk;

        auto cmp = [&](int i, int j) -> bool {
            if (a[i] == a[j]) {
                return i < j;
            }
            if constexpr (IS_MIN) {
                return a[i] < a[j];
            } else {
                return a[j] < a[i];
            }
        };

        for (int i = 0; i < n; i++) {
            while (!stk.empty() && cmp(i, stk.back())) {
                L[i] = stk.back();
                stk.pop_back();
            }
            range[i].first = (stk.empty() ? 0 : stk.back() + 1);
            if (!stk.empty()) {
                R[stk.back()] = i;
            }
            stk.push_back(i);
        }
        if (!stk.empty()) {
            root = stk.front();
        }
        stk.clear();
        for (int i = n - 1; i >= 0; i--) {
            while (!stk.empty() && cmp(i, stk.back())) {
                stk.pop_back();
            }
            range[i].second = stk.empty() ? n : stk.back();
            stk.push_back(i);
        }
```

```
        for (int i = 0; i < n; i++) {
            if (L[i] != -1) {
                p[L[i]] = i;
            }
            if (R[i] != -1) {
                p[R[i]] = i;
            }
        }
    }
};
```

8 线性代数

8.1 矩阵

```
template <typename T>
struct Matrix {
    int n, m;
    vector<vector<T>> a;

    Matrix() : n(0), m(0) {}
    Matrix(int n_, int m_, T v = T{}) : n(n_), m(m_), a(n, vector<T>(m, v)) {}
    Matrix(int n_) : Matrix(n_, n_) {}

    static Matrix identity(int n) {
        Matrix res(n);
        for (int i = 0; i < n; i++) {
            res[i][i] = T{1};
        }
        return res;
    }

    vector<T>& operator[](int i) {
        return a[i];
    }
    const vector<T>& operator[](int i) const {
        return a[i];
    }
};

template <typename T>
Matrix<T> operator*(const Matrix<T>& a, const Matrix<T>& b) {
    assert(a.m == b.n);
    Matrix<T> c(a.n, b.m);
    for (int i = 0; i < c.n; i++) {
        for (int k = 0; k < a.m; k++) {
            for (int j = 0; j < c.m; j++) {
                c.a[i][j] += a.a[i][k] * b.a[k][j];
            }
        }
    }
    return c;
}

template <typename T>
Matrix<T>& operator*=(Matrix<T>& a, const Matrix<T>& b) {
    return a = a * b;
}

template <typename T, integral U>
Matrix<T> power(Matrix<T> a, U b) {
    Matrix<T> res = Matrix<T>::identity(a.n);
    for (; b > 0; b >>= 1) {
        if (b & 1) {
            res *= a;
        }
        a *= a;
    }
    return res;
}

template <typename T>
Matrix<T> transpose(const Matrix<T>& a) {
    Matrix<T> res(a.m, a.n);
    for (int i = 0; i < a.n; i++) {
        for (int j = 0; j < a.m; j++) {
            res[j][i] = a[i][j];
        }
    }
    return res;
}

template <typename T>
Matrix<T> augment(const Matrix<T>& a, const Matrix<T>& b) {
    assert(a.n == b.n);
    Matrix<T> res(a.n, a.m + b.m);
    for (int i = 0; i < a.n; i++) {
        for (int j = 0; j < a.m; j++) {
            res[i][j] = a[i][j];
        }
        for (int j = 0; j < b.m; j++) {
```

```
            res[i][a.m + j] = b[i][j];
        }
    }
    return res;
}

template <typename T>
Matrix<T> augment(const Matrix<T>& a, const vector<T>& b) {
    assert(a.n == int(b.size()));
    Matrix<T> res(a.n, a.m + 1);
    for (int i = 0; i < a.n; i++) {
        for (int j = 0; j < a.m; j++) {
            res[i][j] = a[i][j];
        }
        res[i][a.m] = b[i];
    }
    return res;
}
```

8.2 高斯消元

```
template <typename T>
pair<vector<int>, T> eliminate(Matrix<T>& A, int lim, bool full) {
    // full == true -> RREF
    // full == false -> REF
    vector<int> pivot;
    T prod = T{1};
    int r = 0;
    for (int j = 0; j < lim && r < A.n; j++) {
        int p = r;
        while (p < A.n && A[p][j] == T{0}) {
            p++;
        }
        if (p == A.n) {
            continue;
        }
        if (p != r) {
            swap(A[p], A[r]);
            prod = -prod;
        }
        T x = A[r][j];
        prod *= x;
        pivot.push_back(j);
        T inv = T{1} / x;
        for (int k = j; k < A.m; k++) {
            A[r][k] *= inv;
        }
        for (int i = 0; i < A.n; i++) {
            if (i == r || (!full && i < r)) {
                continue;
            }
            T c = A[i][j];
            if (c == T{0}) {
                continue;
            }
            A[i][j] = T{0};
            for (int k = j + 1; k < A.m; k++) {
                A[i][k] -= c * A[r][k];
            }
        }
        r++;
    }
    return {pivot, prod};
}

template <typename T>
pair<vector<int>, T> gauss(Matrix<T>& A) {
    return eliminate(A, A.m, false);
}
template <typename T>
pair<vector<int>, T> gaussJordan(Matrix<T>& A, int lim) {
    return eliminate(A, lim, true);
}

template <typename T>
pair<T, Matrix<T>> matrixInv(const Matrix<T>& a) {
    if (a.n != a.m) {
        return {T{0}, {}};
    }
    int n = a.n;
    auto aug = augment(a, Matrix<T>::identity(n));
    auto [pivot, det] = gaussJordan(aug, n);
    if (pivot.size() < n) {
        return {T{0}, {}};
    }
    Matrix<T> inv(n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            inv[i][j] = aug[i][n + j];
```

```
        }
    }
    return {det, inv};
}

template <typename T>
int matrixRank(Matrix<T> a) {
    auto [pivot, prod] = gauss(a);
    return pivot.size();
}

template <typename T>
T determinant(Matrix<T> A) {
    assert(A.n == A.m);
    auto [pivot, det] = gauss(A);
    if (pivot.size() < A.n) {
        return T{0};
    }
    return det;
}

template <typename T>
vector<vector<T>> solveLinear(const Matrix<T>& A, const vector<T>& b) {
    int n = A.n;
    int m = A.m;
    auto aug = augment(A, b);
    auto [pivot, prod] = gaussJordan(aug, m);
    int r = pivot.size();
    for (int i = r; i < n; i++) {
        if (aug[i][m] != T{0}) {
            return {};
        }
    }
    vector<int> row(m, -1);
    for (int i = 0; i < r; i++) {
        row[pivot[i]] = i;
    }
    vector<vector<T>> sol(1, vector<T>(m));
    for (int j = 0; j < m; j++) {
        if (row[j] != -1) {
            sol[0][j] = aug[row[j]][m];
        }
    }
    for (int j = 0; j < m; j++) {
        if (row[j] != -1) {
            continue;
        }
        vector<T> x(m);
        x[j] = T{-1};
        for (int i = 0; i < r; i++) {
            x[pivot[i]] = aug[i][j];
        }
        sol.push_back(x);
    }
    return sol;
}
```

8.3 GF(2) 空间

8.3.1 异或线性基

```
struct Basis {
    array<u64, 64> p{};
    bool norm = false;

    int size() {
        int r = 0;
        for (int i = 0; i < 64; i++) {
            r += p[i] != 0;
        }
        return r;
    }

    bool insert(u64 x) {
        for (int i = 63; i >= 0; i--) {
            if (!(x >> i & 1)) {
                continue;
            }
            if (!p[i]) {
                p[i] = x;
                norm = false;
                return true;
            }
            x ^= p[i];
        }
        return false;
    }

    bool contain(u64 x) {
        for (int i = 63; i >= 0; i--) {
            if (x >> i & 1) {
```

```
                x ^= p[i];
            }
        }
        return x == 0;
    }

    u64 getMax(u64 x = 0) {
        for (int i = 63; i >= 0; i--) {
            if ((x ^ p[i]) > x) {
                x ^= p[i];
            }
        }
        return x;
    }

    u64 getMin(u64 x = 0) {
        for (int i = 63; i >= 0; i--) {
            if ((x ^ p[i]) < x) {
                x ^= p[i];
            }
        }
        return x;
    }

    void normalize() {
        if (norm == true) {
            return;
        }
        norm = true;
        for (int i = 63; i >= 0; i--) {
            if (!p[i]) {
                continue;
            }
            for (int j = i - 1; j >= 0; j--) {
                if (p[i] >> j & 1) {
                    p[i] ^= p[j];
                }
            }
        }
    }

    u64 kth(u64 k) {
        normalize();
        vector<u64> b;
        for (int i = 0; i < 64; i++) {
            if (p[i]) {
                b.push_back(p[i]);
            }
        }
        int r = b.size();
        if (r < 64) {
            if (k >= (1ULL << r)) {
                return -1;
            }
        }
        u64 res = 0;
        for (int i = 0; i < r; i++) {
            if (k >> i & 1) {
                res ^= b[i];
            }
        }
        return res;
    }
};
```

8.3.2 逆矩阵

```
namespace gf2 {

template <typename U>
vector<U> inverse(int n, vector<U> a) {
    vector<U> b(n);
    for (int i = 0; i < n; i++) {
        b[i] = U{1} << i;
    }
    for (int j = 0; j < n; j++) {
        int p = j;
        while (p < n && !(a[p] >> j & U{1})) {
            p++;
        }
        if (p == n) {
            return {};
        }
        swap(a[p], a[j]);
        swap(b[p], b[j]);
        for (int i = 0; i < n; i++) {
            if (i != j && (a[i] >> j & U{1})) {
                a[i] ^= a[j];
                b[i] ^= b[j];
            }
```

```
        }
    }
    return b;
}

}
```

8.3.3 转置

```
namespace gf2 {

template <typename U>
vector<U> transpose(int n, int m, const vector<U>& a) {
    vector<U> b(m);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (a[i] >> j & U{1}) {
                b[j] |= U{1} << i;
            }
        }
    }
    return b;
}

}
```

8.3.4 GF(2) 线性方程组

```
namespace gf2 {

template <unsigned_integral U>
vector<U> solveLinear(int n, int m, vector<U> a, U b) {
    vector<U> mat(m);
    U rhs = 0;
    for (int i = 0; i < n; i++) {
        U x = a[i], c = b >> i & 1;
        for (int j = m - 1; j >= 0; j--) {
            if ((x ^ mat[j]) < x) {
                x ^= mat[j];
                c ^= rhs >> j & 1;
            }
        }
        if (!x) {
            if (c) {
                return {};
            }
            continue;
        }
        int k = __lg(x);
        for (int j = k; j < m; j++) {
            if ((mat[j] ^ x) < mat[j]) {
                mat[j] ^= x;
                rhs ^= c << j;
            }
        }
        mat[k] = x;
        rhs |= c << k;
    }
    vector<U> res{rhs};
    mat = transpose(m, m, mat);
    for (int j = 0; j < m; j++) {
        if (!(mat[j] >> j & 1)) {
            res.push_back(mat[j] | (U{1} << j));
        }
    }
    return res;
}

}
```

9 字符串

9.1 Manacher

```
vector<int> manacher(string s) {
    string t = "#";
    for (auto c : s) {
        t += c;
        t += '#';
    }
    int n = t.size();
    vector<int> r(n);
    for (int i = 0, j = 0; i < n; i++) {
        if (2 * j - i >= 0 && j + r[j] > i) {
            r[i] = min(r[2 * j - i], j + r[j] - i);
        }
        while (i - r[i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]]) {
            r[i] += 1;
        }
        if (i + r[i] > j + r[j]) {
            j = i;
        }
    }
```

```
    return r;
}
```

9.2 KMP

```
vector<int> kmp(string s) {
    int n = s.size();
    vector<int> f(n + 1);
    for (int i = 1, j = 0; i < n; i++) {
        while (j && s[i] != s[j]) {
            j = f[j];
        }
        j += (s[i] == s[j]);
        f[i + 1] = j;
    }
    return f;
}

```

9.3 Trie

```
struct Trie {
    struct Node {
        array<int, 26> nxt;
        int pass, end;
        Node() {
            nxt.fill(-1);
            end = 0;
            pass = 0;
        }
    };

    vector<Node> tr;
    int tot;

    Trie(int n) {
        init(n);
    }

    void init(int n) {
        tr.assign(n, {});
        tot = 0;
    }

    void insert(string& s) {
        int p = 0;
        ++tr[p].pass;
        for (char& ch : s) {
            int c = ch - 'a';
            if (tr[p].nxt[c] == -1) {
                ++tot;
                tr[p].nxt[c] = tot;
            }
            p = tr[p].nxt[c];
            ++tr[p].pass;
        }
        ++tr[p].end;
    }

    int count(string& s) {
        int p = 0;
        for (char& ch : s) {
            int c = ch - 'a';
            if (tr[p].nxt[c] == -1) {
                return false;
            }
            p = tr[p].nxt[c];
        }
        return tr[p].end;
    }
};

```

9.4 Z 函数

```
vector<int> Z(string s) {
    int n = s.size();
    vector<int> z(n + 1);
    z[0] = n;
    for (int i = 1, j = 1; i < n; i++) {
        z[i] = max(0, min(j + z[j] - i, z[i - j]));
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if (i + z[i] > j + z[j]) {
            j = i;
        }
    }
    return z;
}

```

9.5 AC 自动机

```
struct AhoCorasick {
    static constexpr int ALPHABET = 26;
    struct Node {
        int len;
        int link;
        array<int, ALPHABET> next;
        Node() : len{0}, link{0}, next{} {}
    };

    vector<Node> t;

    AhoCorasick() {
        init();
    }

    void init() {
        t.assign(2, Node());
        t[0].next.fill(1);
        t[0].len = -1;
    }

    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }

    int add(const string &a) {
        int p = 1;
        for (auto c : a) {
            int x = c - 'a';
            if (t[p].next[x] == 0) {
                t[p].next[x] = newNode();
                t[t[p].next[x]].len = t[p].len + 1;
            }
            p = t[p].next[x];
        }
        return p;
    }

    void work() {
        queue<int> q;
        q.push(1);

        while (!q.empty()) {
            int x = q.front();
            q.pop();

            for (int i = 0; i < ALPHABET; i++) {
                if (t[x].next[i] == 0) {
                    t[x].next[i] = t[t[x].link].next[i];
                } else {
                    t[t[x].next[i]].link = t[t[x].link].next[i];
                    q.push(t[x].next[i]);
                }
            }
        }
    }

    int next(int p, int x) {
        return t[p].next[x];
    }

    int link(int p) {
        return t[p].link;
    }

    int len(int p) {
        return t[p].len;
    }

    int size() {
        return t.size();
    }
};
```

10 随机化

10.1 随机数

```
static mt19937_64
rng(static_cast<u64>(chrono::steady_clock::now().time_since_epoch().count()));

i64 randInt(i64 l, i64 r) {
    assert(l < r);
    return uniform_int_distribution<i64>(l, r - 1)(rng);
}

u64 randInt(u64 n) {
    assert(n > 0);
    return uniform_int_distribution<u64>(0, n - 1)(rng);
```

```
}

bool randBool() {
    return randInt(2);
}

double randReal(double lo, double hi) {
    assert(lo < hi);
    return uniform_real_distribution<double>(lo, hi)(rng);
}
```

10.2 随机数组

```
template <typename T>
void shuffle(vector<T>& a) {
    int n = a.size();
    for (int i = 0; i < n; i++) {
        int j = randInt(0, i + 1);
        if (i != j) {
            swap(a[i], a[j]);
        }
    }
}
// OR ranges::shuffle(a, rng);

template <typename T>
T& randElem(vector<T>& a) {
    assert(!a.empty());
    return a[randInt(a.size())];
}

vector<i64> randArray(int n, i64 l, i64 r) {
    vector<i64> a(n);
    for (auto& x : a) {
        x = randInt(l, r);
    }
    return a;
}

vector<int> randPerm(int n) {
    vector<int> a(n);
    iota(a.begin(), a.end(), 0);
    shuffle(a);
    return a;
}

vector<i64> randDistinct(int n, i64 l, i64 r) {
    unordered_set<i64> s;
    vector<i64> a;
    for (i64 i = r - l - n; i < r - l; i++) {
        i64 x = randInt(i + 1);
        if (!s.insert(x).second) {
            s.insert(i), a.push_back(i + l);
        } else {
            a.push_back(x + l);
        }
    }
    shuffle(a);
    return a;
}
```

10.3 随机图

```
vector<pair<int, int>> randGraph(int n, int m) {
    set<pair<int, int>> s;
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u != v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}

vector<pair<int, int>> randTree(int n) {
    auto p = randPerm(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) {
        e.emplace_back(p[randInt(i)], p[i]);
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randChain(int n) {
    auto p = randPerm(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) {
        e.emplace_back(p[i - 1], p[i]);
    }
```

```
    shuffle(e);
    return e;
}

vector<pair<int, int>> randStar(int n) {
    auto p = randPerm(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) {
        e.emplace_back(p[0], p[i]);
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randConnectedGraph(int n, int m) {
    set<pair<int, int>> s;
    for (auto [u, v] : randTree(n)) {
        s.emplace(min(u, v), max(u, v));
    }
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u > v) {
            swap(u, v);
        }
        if (u != v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    for (auto& [u, v] : e) {
        if (randBool()) {
            swap(u, v);
        }
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randDAG(int n, int m) {
    auto p = randPerm(n);
    set<pair<int, int>> s;
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u < v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e;
    for (auto [u, v] : s) {
        e.emplace_back(p[u], p[v]);
    }
    shuffle(e);
    return e;
}

vector<pair<int, int>> randBipartiteGraph(int n, int k, int m) {
    set<pair<int, int>> s;
    while (m > int(s.size())) {
        s.emplace(randInt(n), n + randInt(k));
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}

vector<pair<int, int>> randSCC(int n, int m) {
    auto p = randPerm(n);
    set<pair<int, int>> s;
    if (n > 1)
        for (int i = 0; i < n; i++)
            s.emplace(p[i], p[(i + 1) % n]);
    while (m > int(s.size())) {
        int u = randInt(n), v = randInt(n);
        if (u != v) {
            s.emplace(u, v);
        }
    }
    vector<pair<int, int>> e(s.begin(), s.end());
    shuffle(e);
    return e;
}
```

10.4 模拟退火

```
// f 越大越好，g(cur, t) 返回新状态，可用温度 t 控制扰动。
template <typename T, typename F, typename G>
T simulatedAnnealing(T cur, F&& f, G&& g, double t0, double t1, double sec) {
    assert(t0 > t1 && t1 > 0 && sec > 0);
    auto beg = chrono::steady_clock::now();
    T ans = cur;
```

```
    auto val = f(cur), res = val;
    while (true) {
        double p = chrono::duration<double>(chrono::steady_clock::now() -
beg).count() / sec;
        if (p >= 1) {
            break;
        }
        double t = exp(lerp(log(t0), log(t1), p));
        T nxt = g(cur, t);
        auto v = f(nxt);
        double d = double(v) - double(val);
        if (d >= 0 || randReal(0, 1) < exp(d / t)) {
            cur = move(nxt);
            val = v;
        }
        if (val > res) {
            ans = cur;
            res = val;
        }
    }
    return ans;
}
```

11 杂项

11.1 二分答案

```
template <typename T, typename F>
T firstTrue(T lo, T hi, F&& pred) {
    while (lo < hi) {
        T mid = (lo + hi) / 2;
        if (pred(mid)) {
            hi = mid;
        } else {
            lo = mid + 1;
        }
    }
    return lo;
}

template <typename T, typename F>
T lastTrue(T lo, T hi, F&& pred) {
    while (lo < hi) {
        T mid = (lo + hi + 1) / 2;
        if (pred(mid)) {
            lo = mid;
        } else {
            hi = mid - 1;
        }
    }
    return lo;
}
```

11.2 三分

```
template <typename T, typename F, typename C = less<>>
T ternarySearch(T lo, T hi, F&& f, C&& comp = {}) {
    for (int i = 0; i < 100; i++) {
        T m1 = lo + (hi - lo) / 3;
        T m2 = hi - (hi - lo) / 3;
        if (comp(f(m1), f(m2))) {
            hi = m2;
        } else {
            lo = m1;
        }
    }
    return (lo + hi) / 2;
}
```

11.3 i128

```
using i128 = __int128;

ostream& operator<<(ostream& os, i128 n) {
    if (n == 0) {
        return os << 0;
    }
    bool neg = n < 0;
    u128 x = neg ? u128(0) - u128(n) : u128(n);
    string s;
    while (x > 0) {
        s += char('0' + x % 10);
        x /= 10;
    }
    if (neg) s += '-';
    reverse(s.begin(), s.end());
    return os << s;
}

i128 toi128(const string& s) {
    bool neg = !s.empty() && s[0] == '-';
    int start = !s.empty() && (s[0] == '-' || s[0] == '+');
```

```
    i128 n = 0;
    for (int i = start; i < int(s.size()); i++) {
        int d = s[i] - '0';
        n = neg ? n * 10 - d : n * 10 + d;
    }
    return n;
}

i128 sqrti128(i128 n) {
    assert(n >= 0);
    i128 lo = 0, hi = i128(1) << 64;
    while (lo < hi) {
        i128 x = (lo + hi + 1) / 2;
        if (x <= n / x) {
            lo = x;
        } else {
            hi = x - 1;
        }
    }
    return lo;
}

i128 gcd(i128 a, i128 b) {
    while (b) {
        a %= b;
        swap(a, b);
    }
    return a;
}
```

11.4 日期

```
const int d[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

bool isLeap(int y) {
    return y % 400 == 0 || (y % 4 == 0 && y % 100 != 0);
}

int daysInMonth(int y, int m) {
    return d[m - 1] + (isLeap(y) && m == 2);
}

int getDay(int y, int m, int d) {
    int ans = 0;
    for (int i = 1970; i < y; i++) {
        ans += 365 + isLeap(i);
    }
    for (int i = 1; i < m; i++) {
        ans += daysInMonth(y, i);
    }
    ans += d;
    return (ans + 2) % 7 + 1;
}
```

11.5 template

```
using namespace std;

using i64 = int64_t;
using u64 = uint64_t;
using u32 = uint32_t;

using i128 = __int128_t;
using u128 = __uint128_t;

template <typename T>
constexpr T inf = numeric_limits<T>::max() / 2;
template <>
constexpr int inf<int> = 1e9;
template <>
constexpr i64 inf<i64> = 1e18;

void solve();

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t;
    cin >> t;

    while (t--) {
        solve();
    }

    return 0;
}
```

11.6 debug

```
using namespace std;

#define dbg(...) do { cerr << "[" << __LINE__ << "] "; pr(__VA_ARGS__); cerr << '\n'; } while (0)
```

```

template <class T, class... U>
void pr(T x, U... y) {
    if constexpr (requires { cerr << x; }) {
        cerr << x;
    } else if constexpr (requires { tuple_size<T>::value; }) {
        cerr << '(';
        bool f = 0;
        apply([&](auto... e) { ((cerr << (exchange(f, 1) ? "," : ""),
pr(e)), ...); }, x);
        cerr << ')';
    } else {
        cerr << '[';
        bool f = 0;
        for (auto e : x) cerr << (exchange(f, 1) ? "," : ""), pr(e);
        cerr << ']';
    }
    if constexpr (sizeof...(y)) cerr << "  ", pr(y...);
}
```

11.7 对拍

```
#!/bin/bash
# bash stress.sh ./sol ./brute ./gen
set -e
for ((i = 1; ; i++)); do
    "$3" > in.txt
    "$1" < in.txt > out.txt
    "$2" < in.txt > ans.txt
    if ! diff -w out.txt ans.txt > diff.txt; then
        printf '\nWA %d\n' "$i"
        break
    fi
    printf '\rAC %d' "$i"
done
```

11.8 对拍(with SPJ)

```
#!/bin/bash
# bash stress_spj.sh ./sol ./spj ./gen
set -e
for ((i = 1; ; i++)); do
    "$3" > in.txt
    "$1" < in.txt > out.txt
    if ! "$2" in.txt out.txt; then
        printf '\nWA %d\n' "$i"
        break
    fi
    printf '\rAC %d' "$i"
done
```

11.9 SPJ

```
using namespace std;

int main(int argc, char* argv[]) {
    ifstream in(argv[1]), out(argv[2]);
    return 0;
}
```